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SECR1013 DIGITAL LOGIC

MODULE 7: LATCHES & FLIP-FLOPS

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Objectives:

- To introduce the various types of latches and flip-flop;
- To explain the characteristic of latches and flip-flop;
- To illustrate the timing diagram associated with each latches and flip-flop operations.

Topics Outline:

LATCH

- S-R Latch
- Gated S-R Latch
- Gated D Latch

FLIP-FLOPS

- S-R Flip-Flop
- J-K Flip-Flop
- D Flip-Flop
- T Flip-Flop



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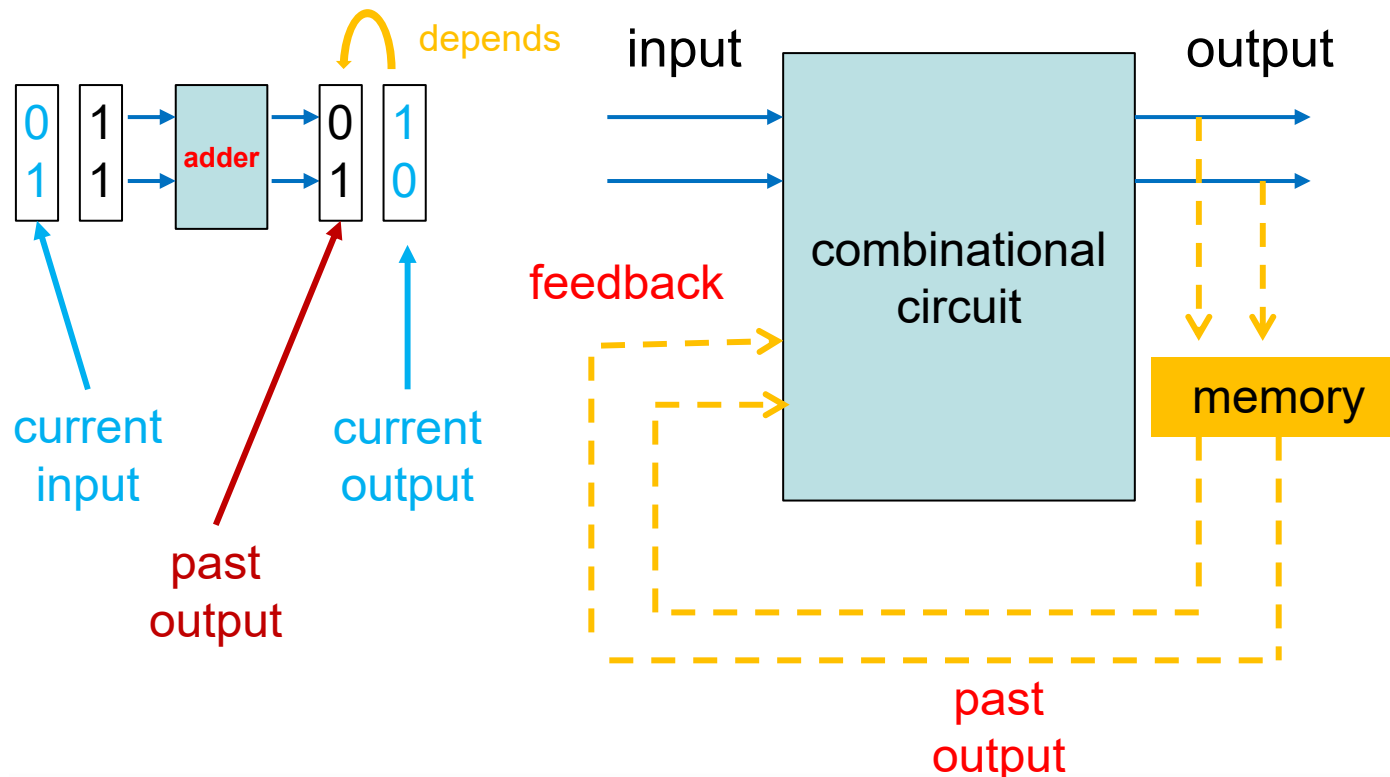
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Introduction to Latch & Flip-Flop



Introduction

- In **sequential circuits**, the current output depends on the current input as well as past output / outputs



Example of seq. circuits:
a) flip flop
b) counter
c) register

counter
0 to 5

0: 0
1: 0 + 1
2: 1 + 1
3: 2 + 1
4: 3 + 1
5: 4 + 1

Sequential Vs Combinational Logic Circuit

unlock:

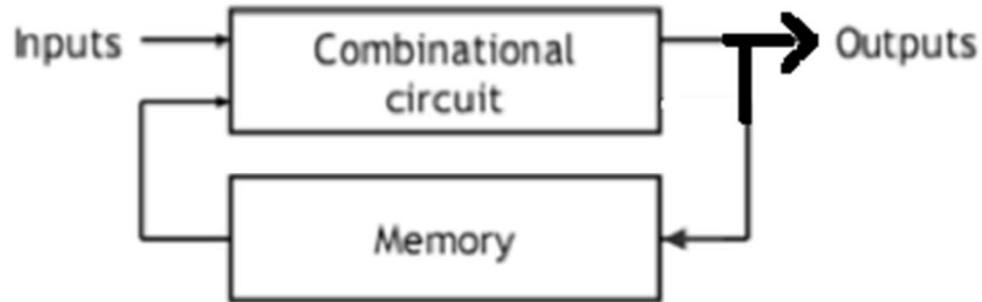
Combinational Logic Circuit:



key sequence is NOT matter

- The output depend on the input
- It has no memory element
 - Therefore, it cannot memorize the previous output

Sequential Logic Circuit:



unlock:

4 → 5 → 6 → 7



To unlock:
key sequence is MATTER

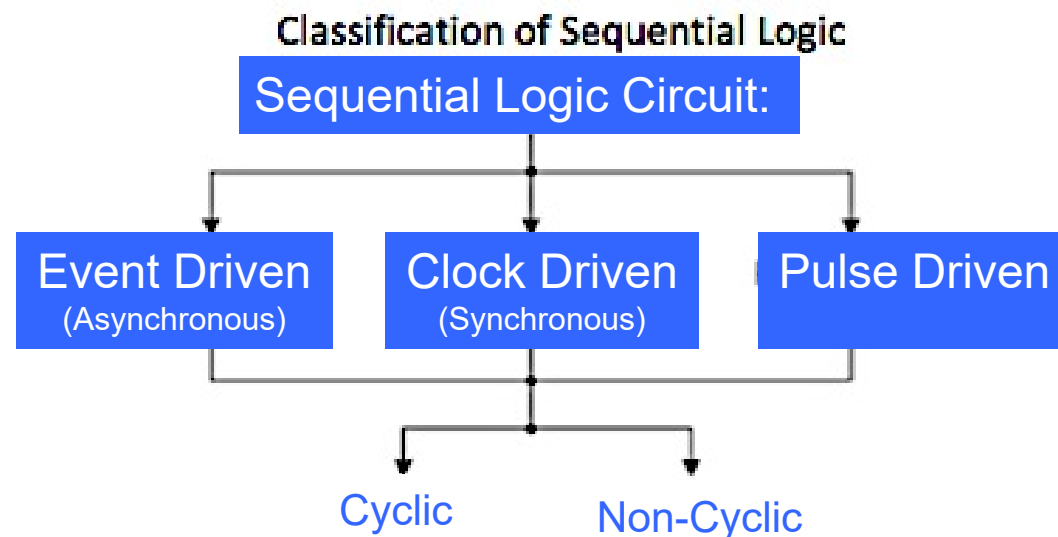
- It has memory element
- The output depend on the input, as well as the previous output from the feedback elements

Sequential Logic circuits can be divided into 3 main categories:

1. Clock Driven - Synchronous Circuits that are synchronized to a specific clock signal.
2. Event Driven - Asynchronous Circuits that react or change state when an external event occurs.
3. Pulse Driven - Which is a Combination of Synchronous and Asynchronous.

→ Bab 8b
↓

Bab 8a



Sequential logic circuits that return back to their original state once reset, i.e. circuits with loops or feedback paths are said to be "Cyclic" in nature.



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Latches

Selak

Latch

- **Latch** is a type of temporary storage device
- It has **two stable** states (bi-stable).
- Latch is level sensitive, or level-triggered.
 - are dependent on the voltage level applied (0/1), not on any signal transition.
- Type of Latch :
 - i) **S-R** *Set - Reset*
 - ii) **Gated S-R**
 - iii) **Gated D** *Data*

LOW

HIGH

INFO :

A **bi-stable** state is one with *two-stable* output states.

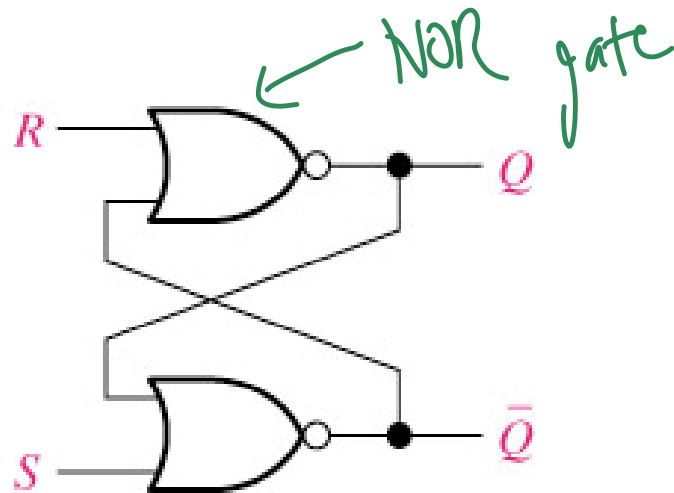
S → Set

R → Reset

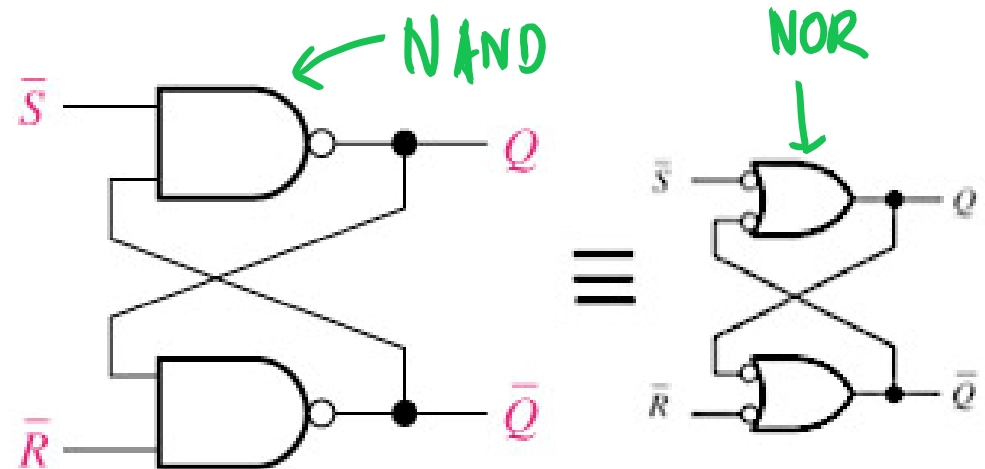
D → Data

(i) S-R Latch

- The output of each gate is connected to an input of the opposite data.
- This produces a **regenerative feedback**.



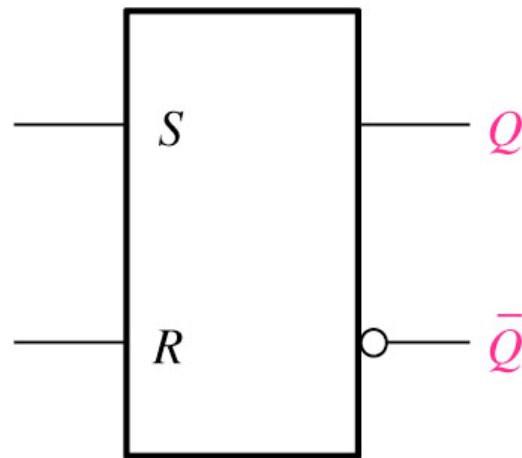
(a) Active-HIGH input S-R latch



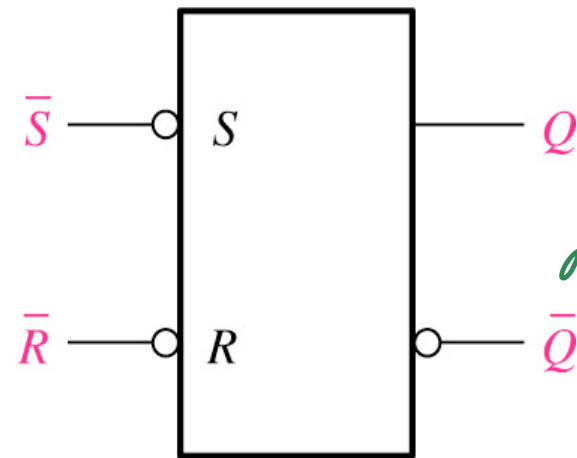
(b) Active-LOW input $\bar{S}$ - $\bar{R}$ latch

↳ Ada bar & terbalik

S-R Latch: Logic symbol



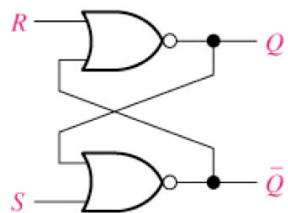
(a) Active-HIGH input S-R latch



(b) Active-LOW input $\bar{S}$ - $\bar{R}$ latch

active-high
NOR

Circuit (a) Active-HIGH input S-R latch :

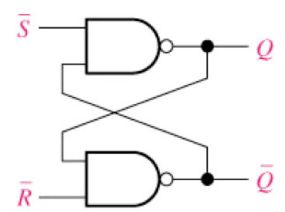


- use **NOR** gate
- Input R and S
- Output Q and Q'

active-low
NAND

Circuit (b) Active-LOW input $\bar{S}$ '- $\bar{R}$ ' latch :

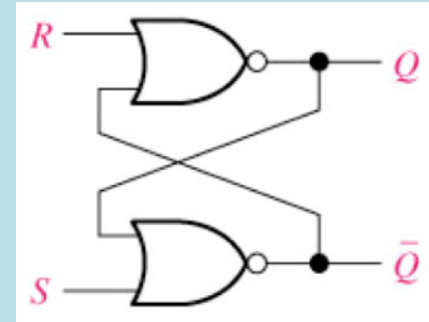
- use **NAND** gate
- Input $\bar{S}$ ' and $\bar{R}$ '
- Output Q and Q'



Active-HIGH

INPUTS		OUTPUTS		COMMENTS
S	R	Q	$\bar{Q}$	
0	0	NC	NC	No change. Latch remains in present state
0	1	0	1	Latch RESET
1	0	1	0	Latch SET
1	1	0	0	Invalid condition

shb dual
set & reset



process

$R = 0 \quad S = 0 \quad Q = 0 \quad Q' = 1$

$R = 0 \quad S = 1 \quad Q = 1 \quad Q' = 0$

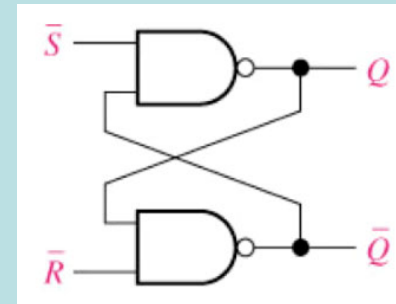
$R = 0 \quad S = 0 \quad Q = 1 \quad Q' = 0$

Explanation :

- The Q and Q -bar outputs are supposed to be in **opposite** states.
- $Q=1$ and Q -bar=0 is defined as **Set** (by making $S=1$ and $R=0$)
- $Q=0$ and Q -bar=1 is defined as **Reset** (by making $S=0$ and $R=1$)
- When S and R are both equal to **0**, the latch's outputs **Not Change** in their prior states.
- If Q and Q -bar happen to be forced to the same state (both 0 or both 1), that state is referred to as **invalid**.

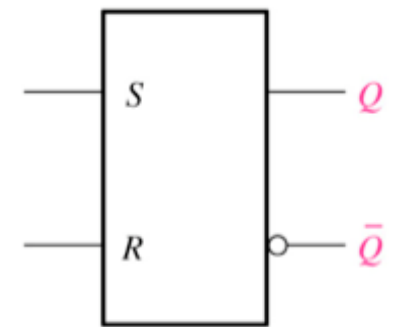
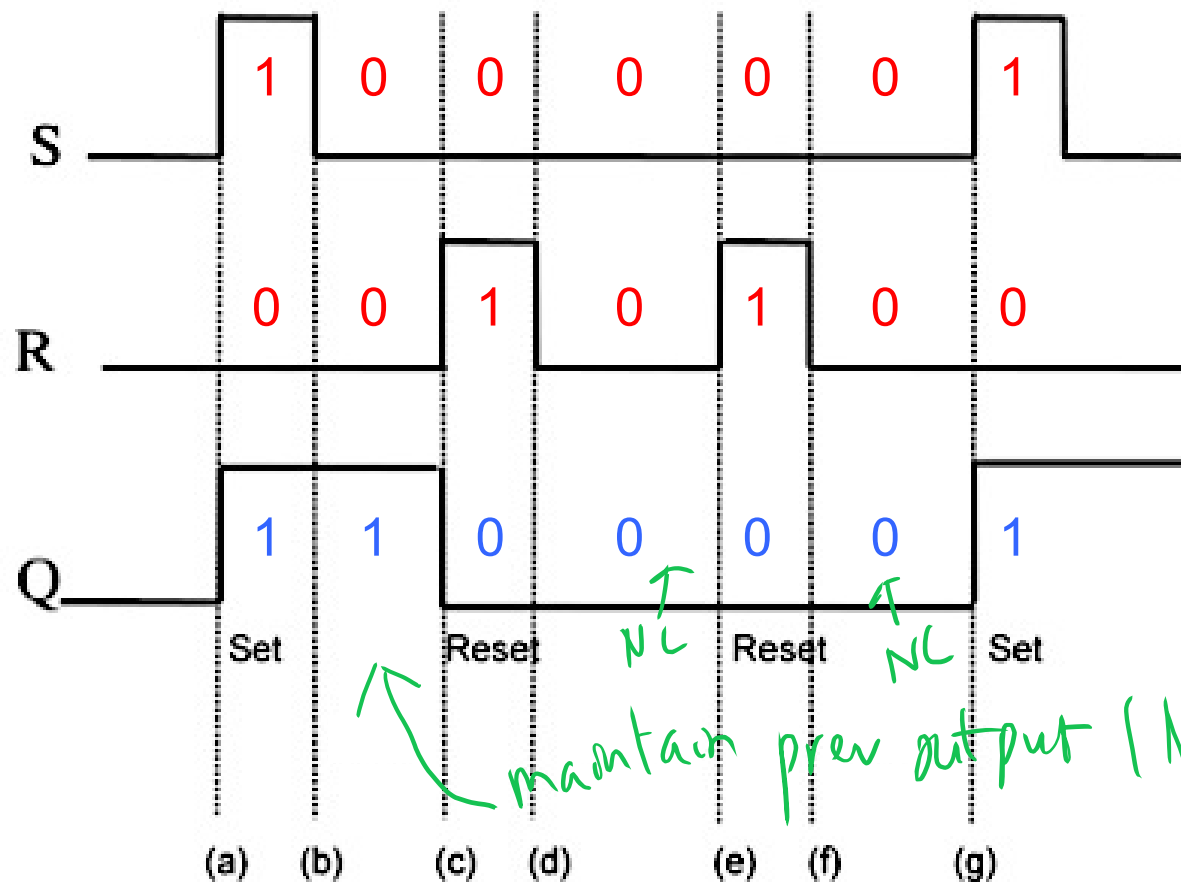
Active-LOW

INPUTS		OUTPUTS		COMMENTS
$\overline{S}$	$\overline{R}$	Q	$\overline{Q}$	
0	0	1	1	Invalid condition
0	1	1	0	Latch SET
1	0	0	1	Latch RESET
1	1	NC	NC	No change. Latch remains in present state



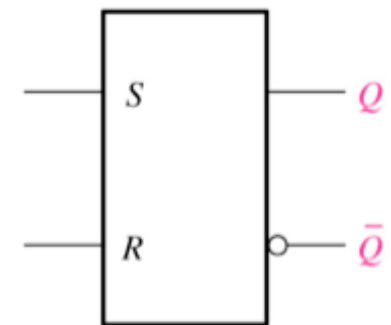
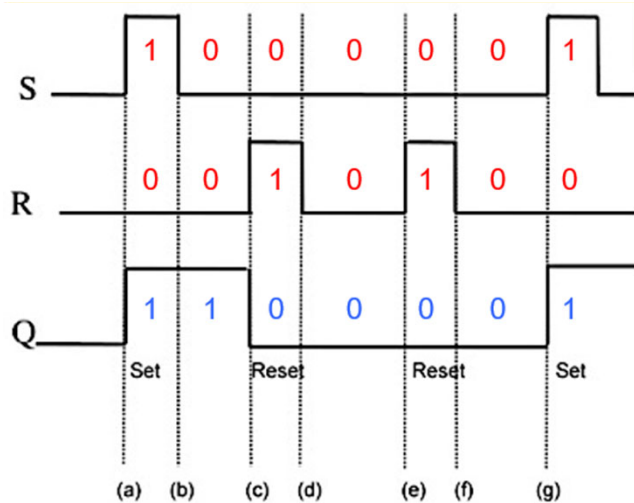
ferbalik je from active-high

Example: S-R Latch (Active HIGH)



Logic symbol

continue...



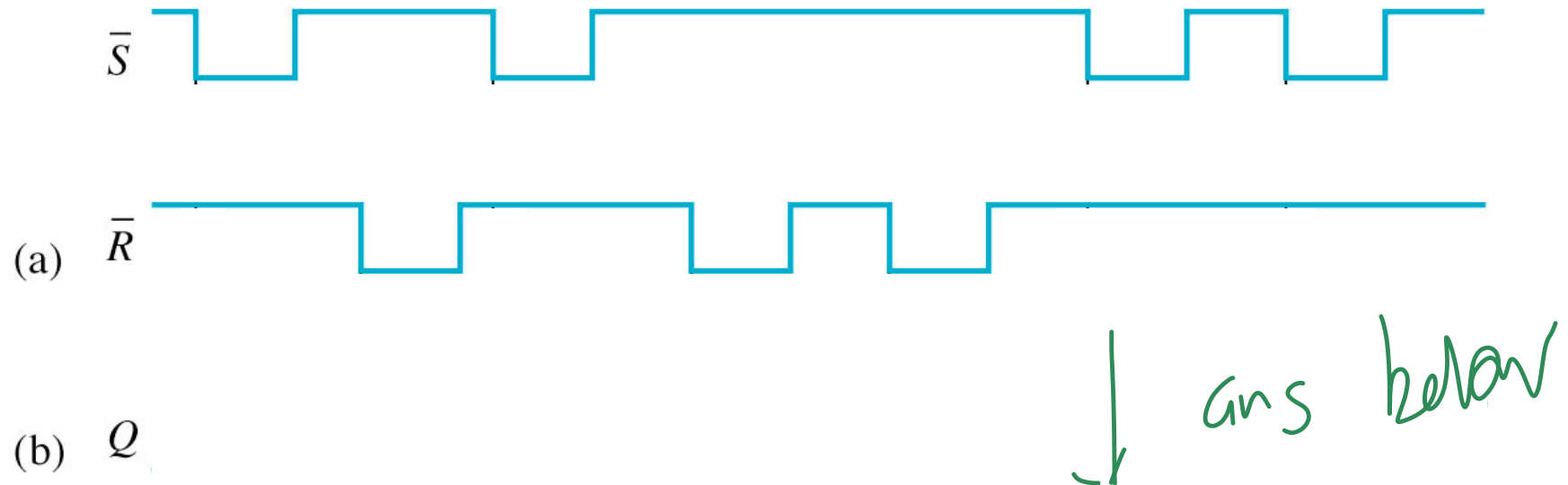
Logic symbol

Explanation:

Location	S	R	Q	State
(a)	1	0	1	SET
(b)	0	0	1	HOLD
(c)	0	1	0	RESET
(d)	0	0	0	HOLD
(e)	0	1	0	RESET
(f)	0	0	0	HOLD
(g)	1	0	1	SET

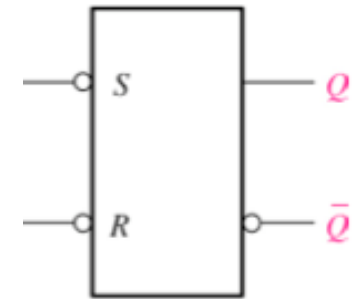
Extra

Exercise 7.1: If the $\bar{S}$ and $\bar{R}$ waveform in (a) are applied to the inputs of latch (active-LOW), determine the waveform that will be observed on the Q output in (b). Assume that Q is initially LOW.



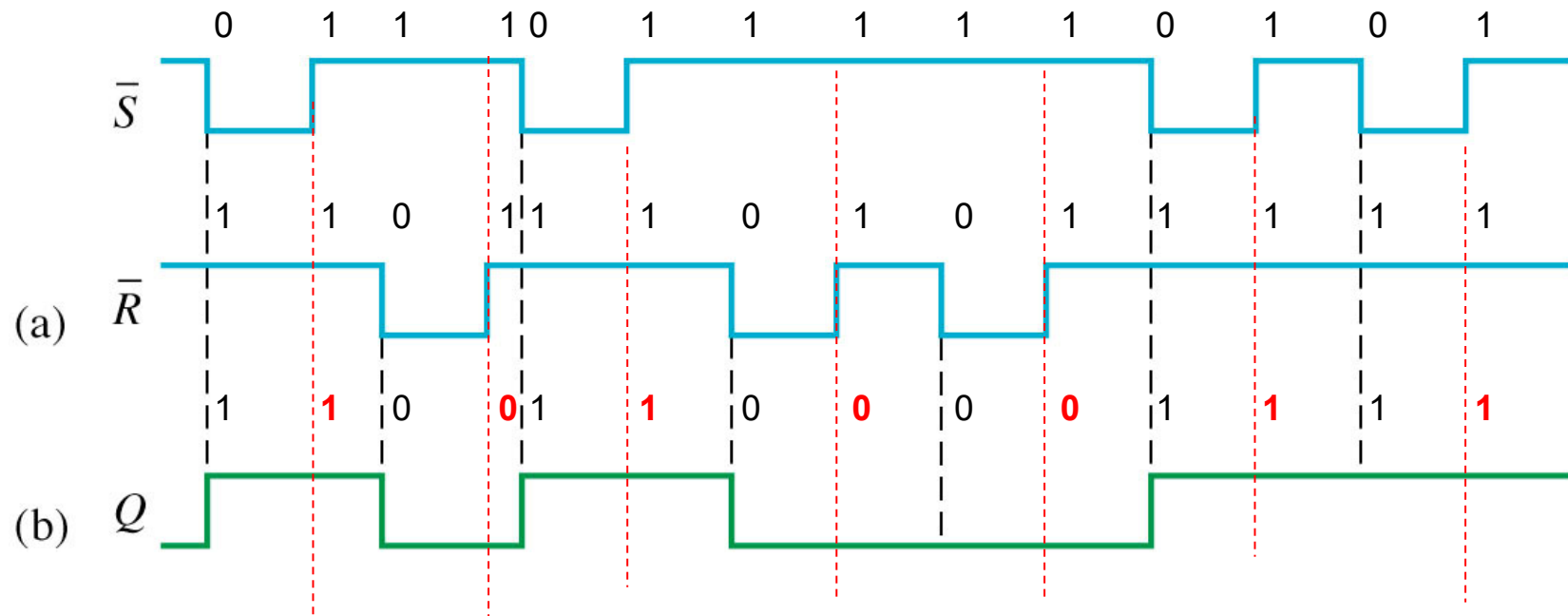
Active-LOW

INPUTS		OUTPUTS		COMMENTS
$\overline{S}$	$\overline{R}$	Q	$\overline{Q}$	
0	0	1	1	Invalid condition
0	1	1	0	Latch SET
1	0	0	1	Latch RESET
1	1	NC	NC	No change. Latch remains in present state



Logic symbol

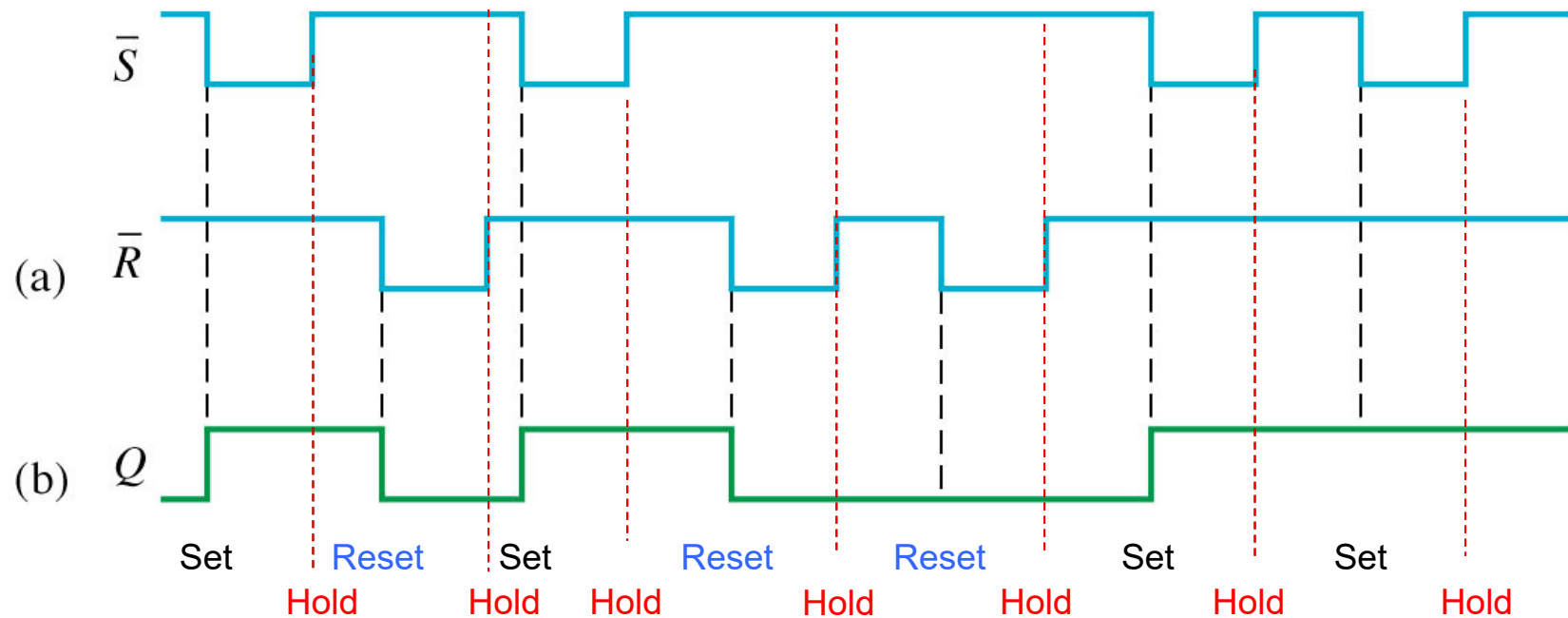
Solution: (Red 0/1 is no change or hold)



Extra

Self-Test:

Define when the states are *set*, *reset*, *hold* or *invalid* from the output waveform for each of the inputs.

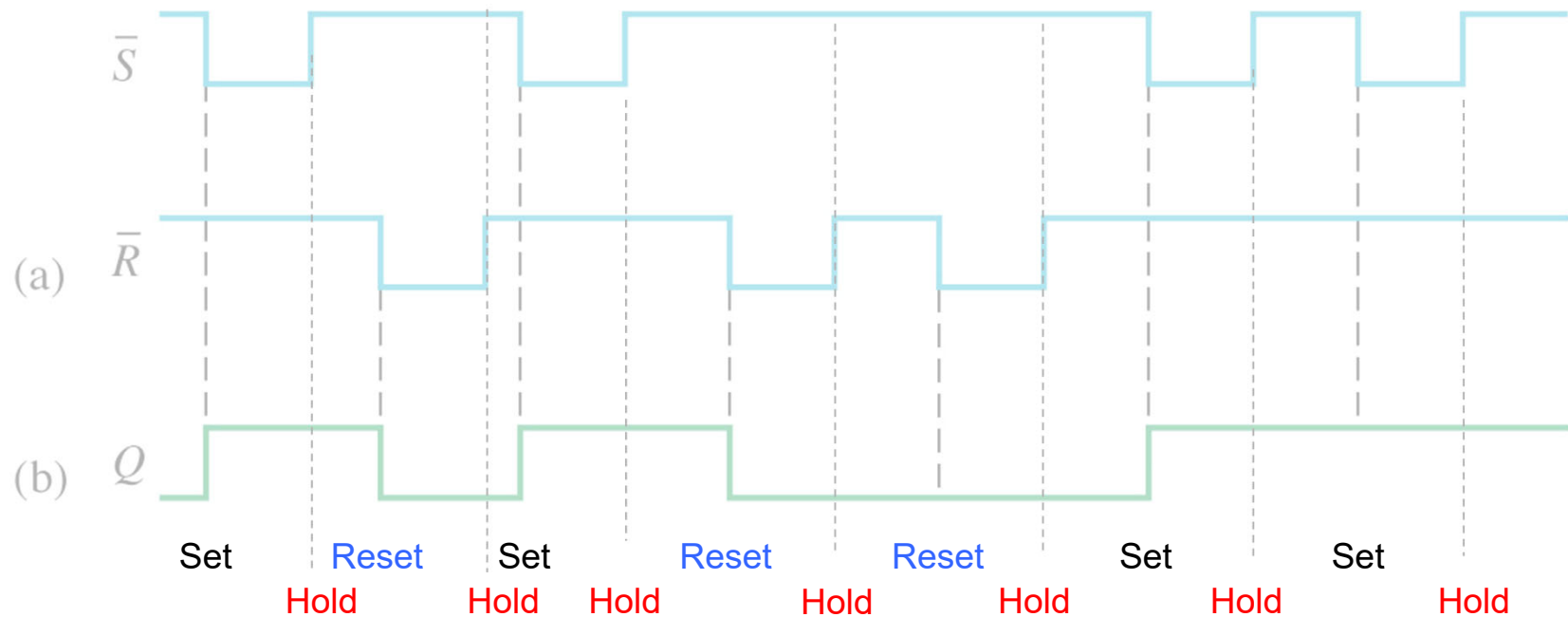


Extra

Self-Test:

Write the sequence of the states (*set*, *reset*, *hold* or *invalid*) for the output Q waveform for all inputs from the beginning.

Set → Hold → Reset → Hold → Set → Hold → Reset → Hold → Reset → Hold → Set → Hold → Set → Hold



(ii) Gated S-R Latch

- Is a SR Latch controlled by an **enable input** (EN) switch.
- The S and R inputs control the state to which the latch will function when a **HIGH** level is **applied** to the **EN** input.
- The latch will **not change** until **EN** is **HIGH**, and as long as it remains HIGH, the output is controlled by the state of the S and R input.
- The **invalid state** occurs when **both** S and R are simultaneously **HIGH**.

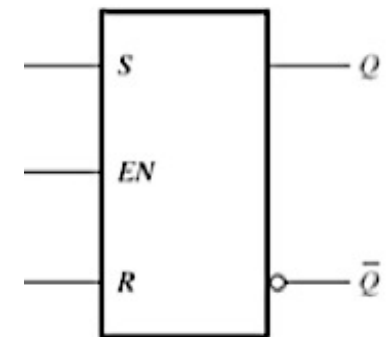
INFO :

EN = 1, Latch is On
EN = 0, Latch is Off

main
switch

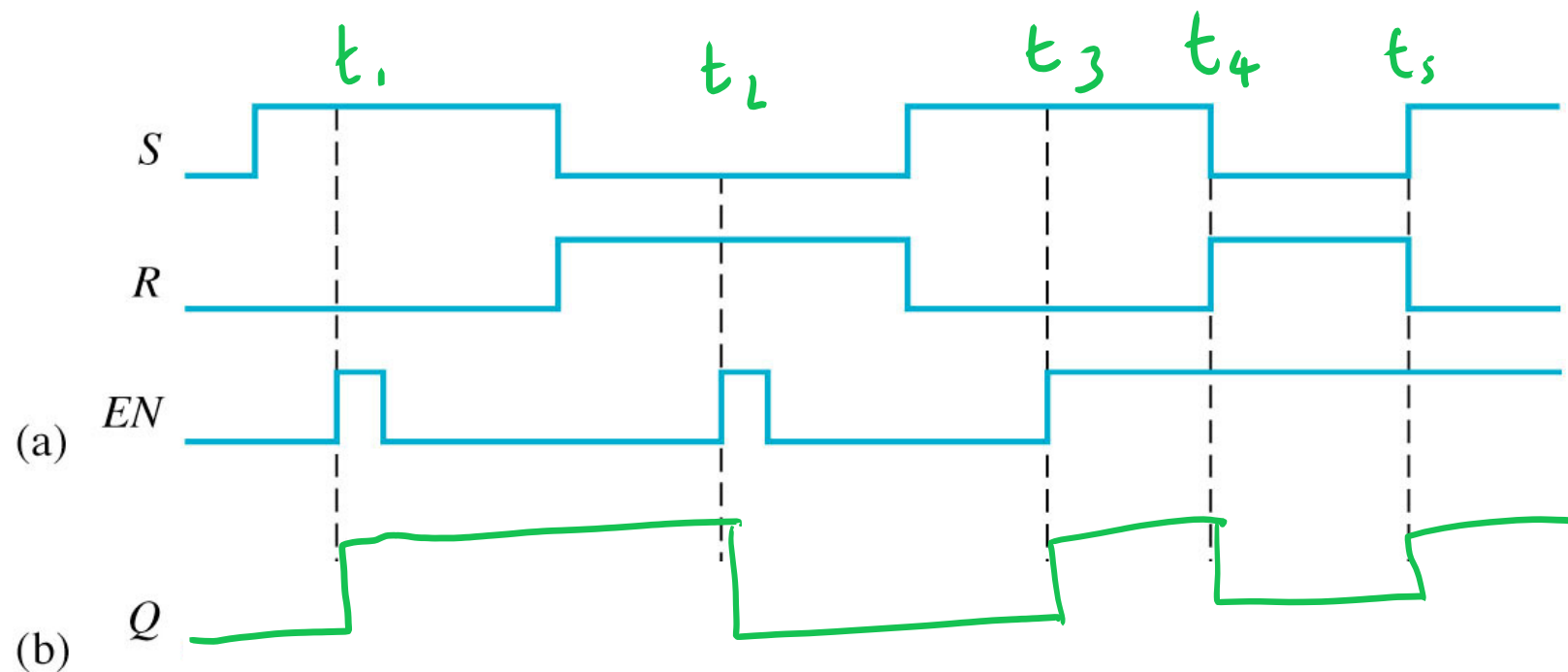
Gated S-R Latch: Truth Table

EN	S	R	Output, Q
0	0/1	0/1	No Change
1	0	0	No Change
1	0	1	Q=0 (Reset)
1	1	0	Q=1 (Set)
1	1	1	Invalid



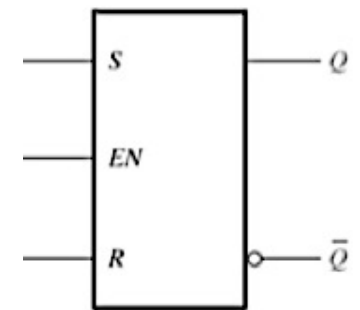
Logic symbol

Example: Gated S-R Latch.
Find the waveform for Q.
Assume that Q is initially LOW.

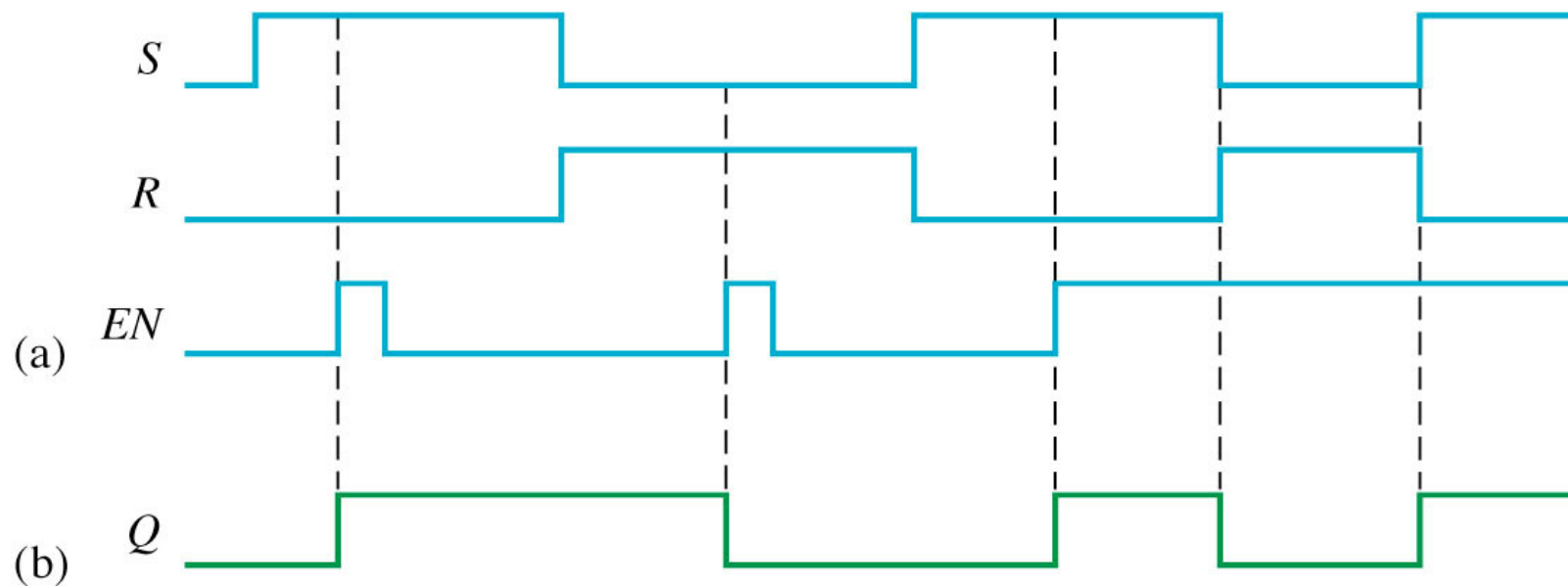


INPUTS		OUTPUTS	
S	R	Q	$\bar{Q}$
0	0	NC	NC
0	1	0	1
1	0	1	0
1	1	0	0

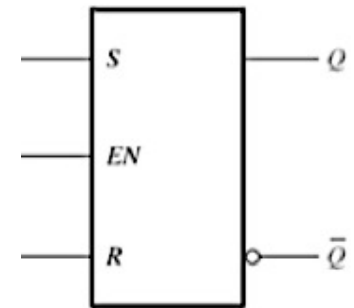
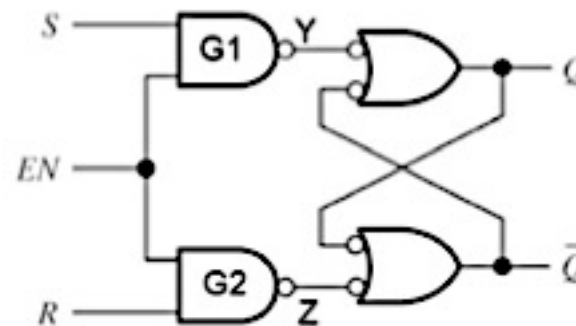
Solution: Assume that Q is initially LOW.



Logic symbol



Exercise 7.2:



Referring to logic diagram above, fill in the table. Assume initially $Q = 0$ and $Q' = 1$

		$\overline{S \cdot EN}$		$\overline{R \cdot EN}$		
EN	S	R	Y	Z	Q	
0	0	0	1	1	0	
0	0	1	1	1	0	
0	1	0	1	1	0	
0	1	1	1	1	0	
1	0	0	1	1	0	
1	0	1	1	0	1	
1	1	0	0	1	0	
1	1	1	0	0	—	

EN off

Set
Reset
Invalid

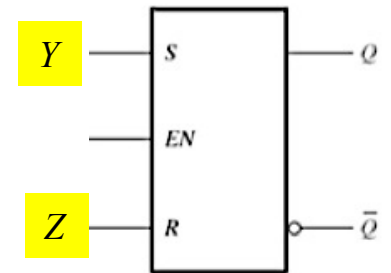
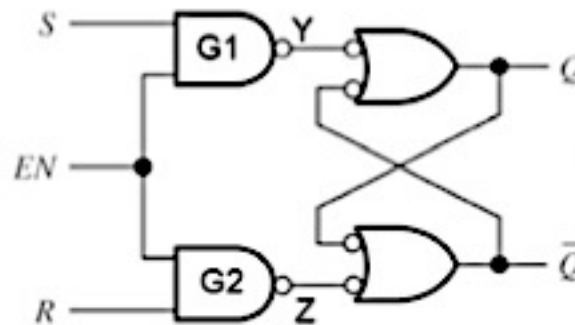
EN on

Active-LOW

INPUTS		OUTPUTS		COMMENTS
$\bar{S}$	$\bar{R}$	Q	$\bar{Q}$	
0	0	1	1	Invalid condition
0	1	1	0	Latch SET
1	0	0	1	Latch RESET
1	1	NC	NC	No change. Latch remains in present state



A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0



Solution:

Referring to logic diagram above, fill in the table. Assume initially $Q = 0$ and $Q' = 1$

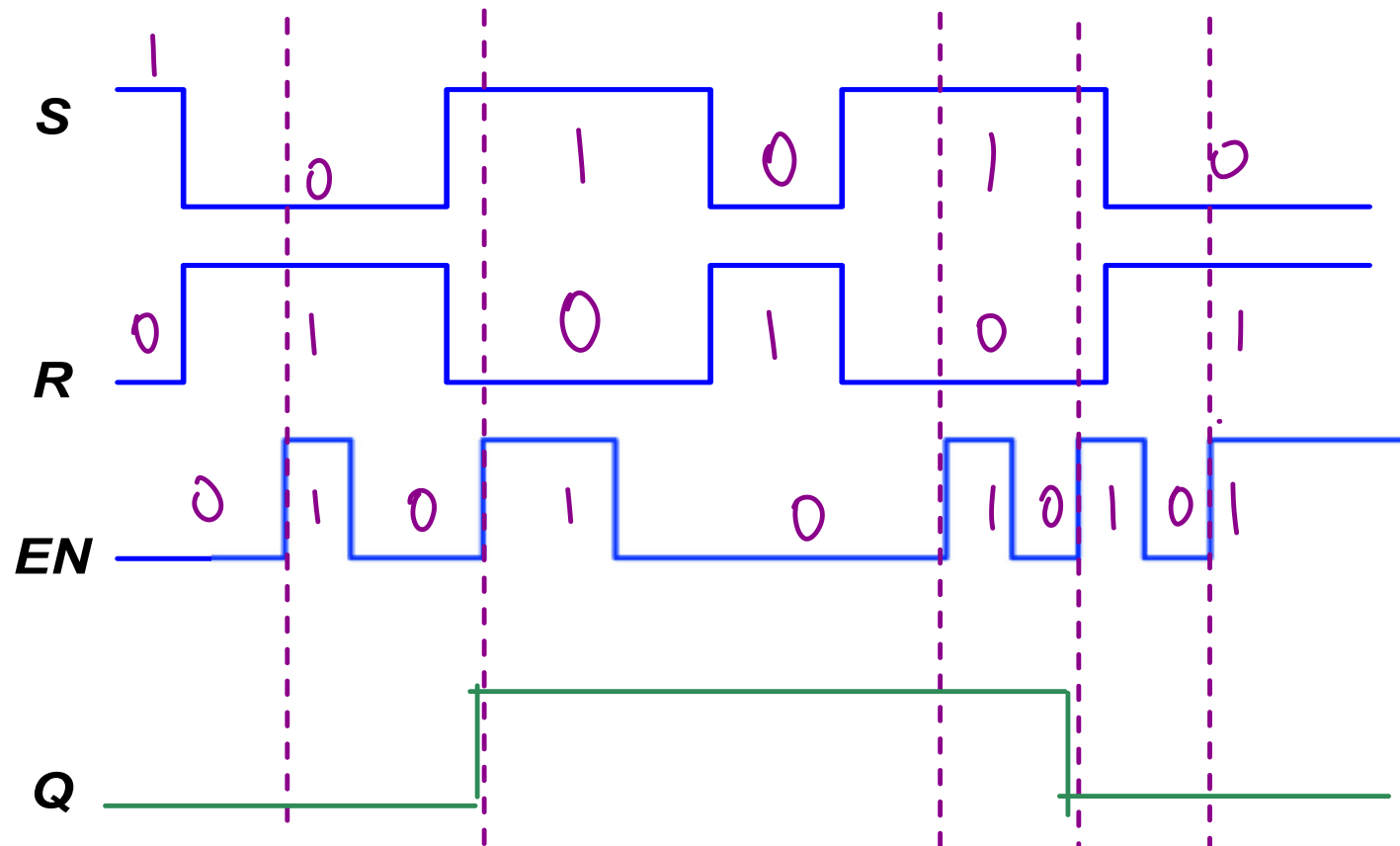
$Q \rightarrow$ Run as 'Active Low S-R' latch when $EN = 1$

		$\overline{S \cdot EN}$		$\overline{R \cdot EN}$		STATE
EN	S	R	Y	Z	Q	
0	0	0	1	1	0	Hold
0	0	1	1	1	0	Hold
0	1	0	1	1	0	Hold
0	1	1	1	1	0	Hold
1	0	0	1	1	0	Hold
1	0	1	1	0	1	Set
1	1	0	0	1	0	Reset
1	1	1	0	0	-	Invalid

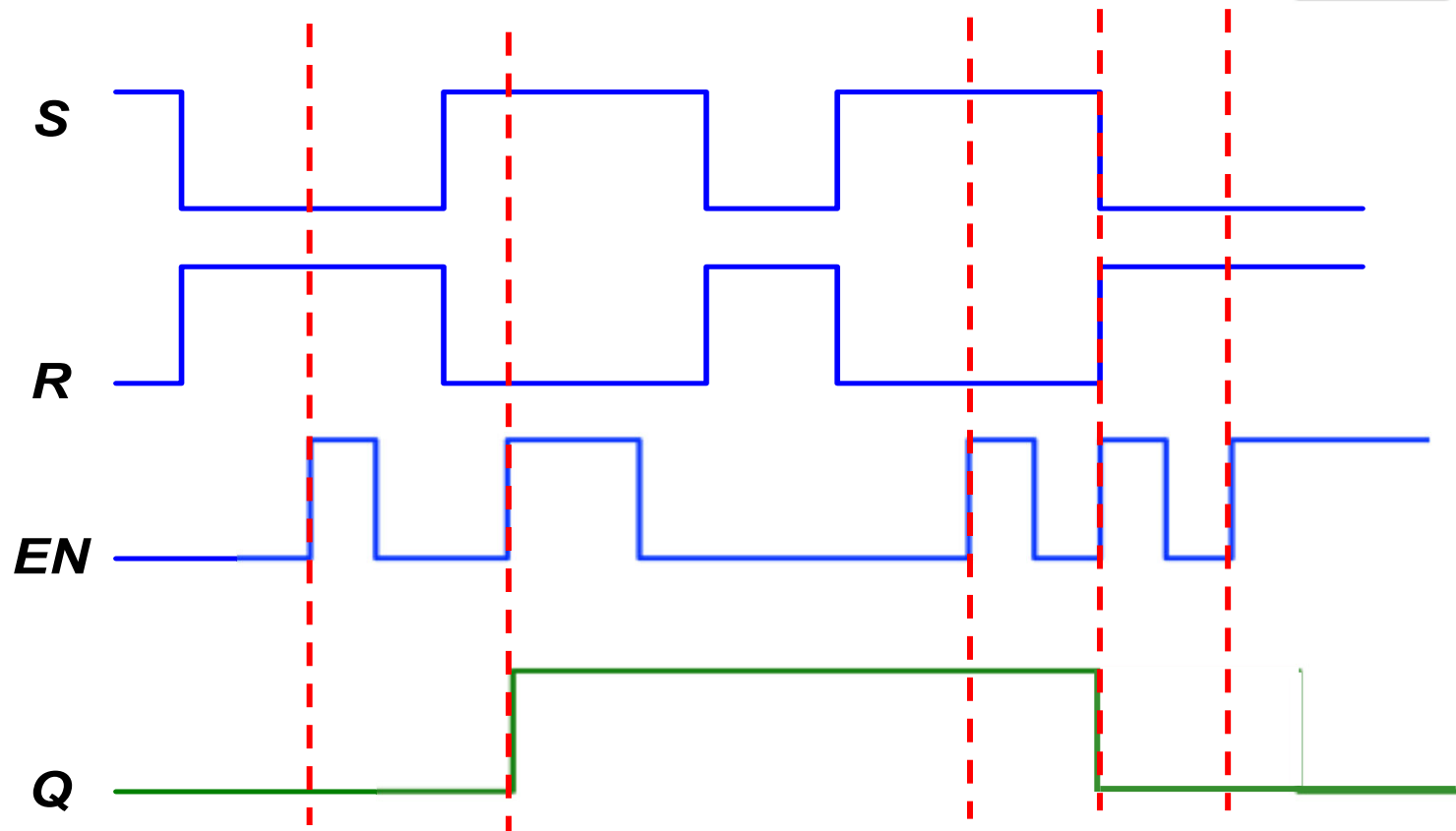
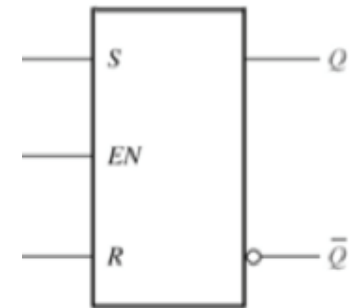
$EN=0$

$EN=1$

Example: Gated S-R Latch.
Find the waveform for Q.
Assume that Q is initially LOW.

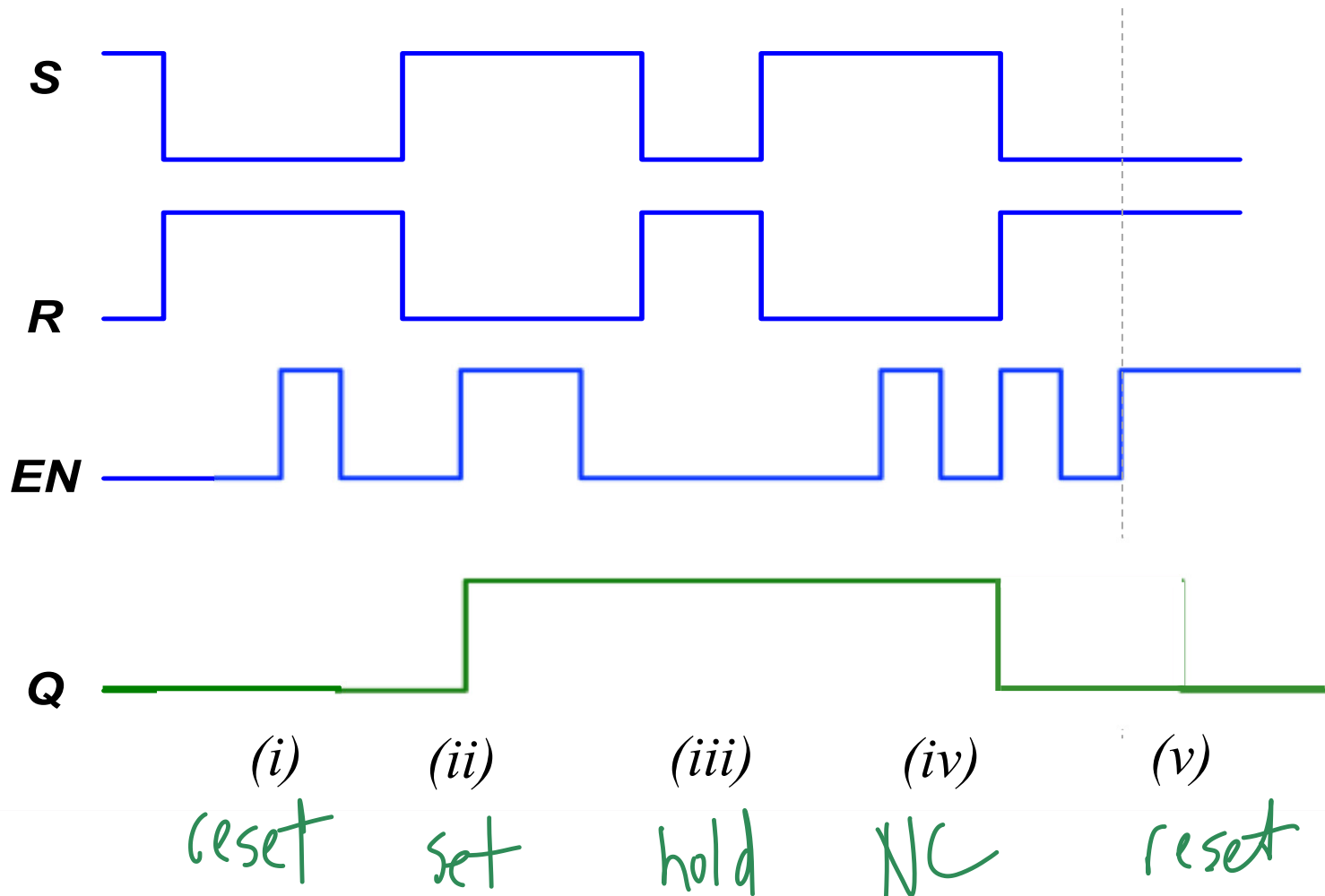


Solution: Assume that Q is initially LOW.



Self-Test:

Define the states (*set*, *reset*, *hold* or *invalid*) for all outputs waveform Q that label as i , ii , iii , iv and v .



Solution:

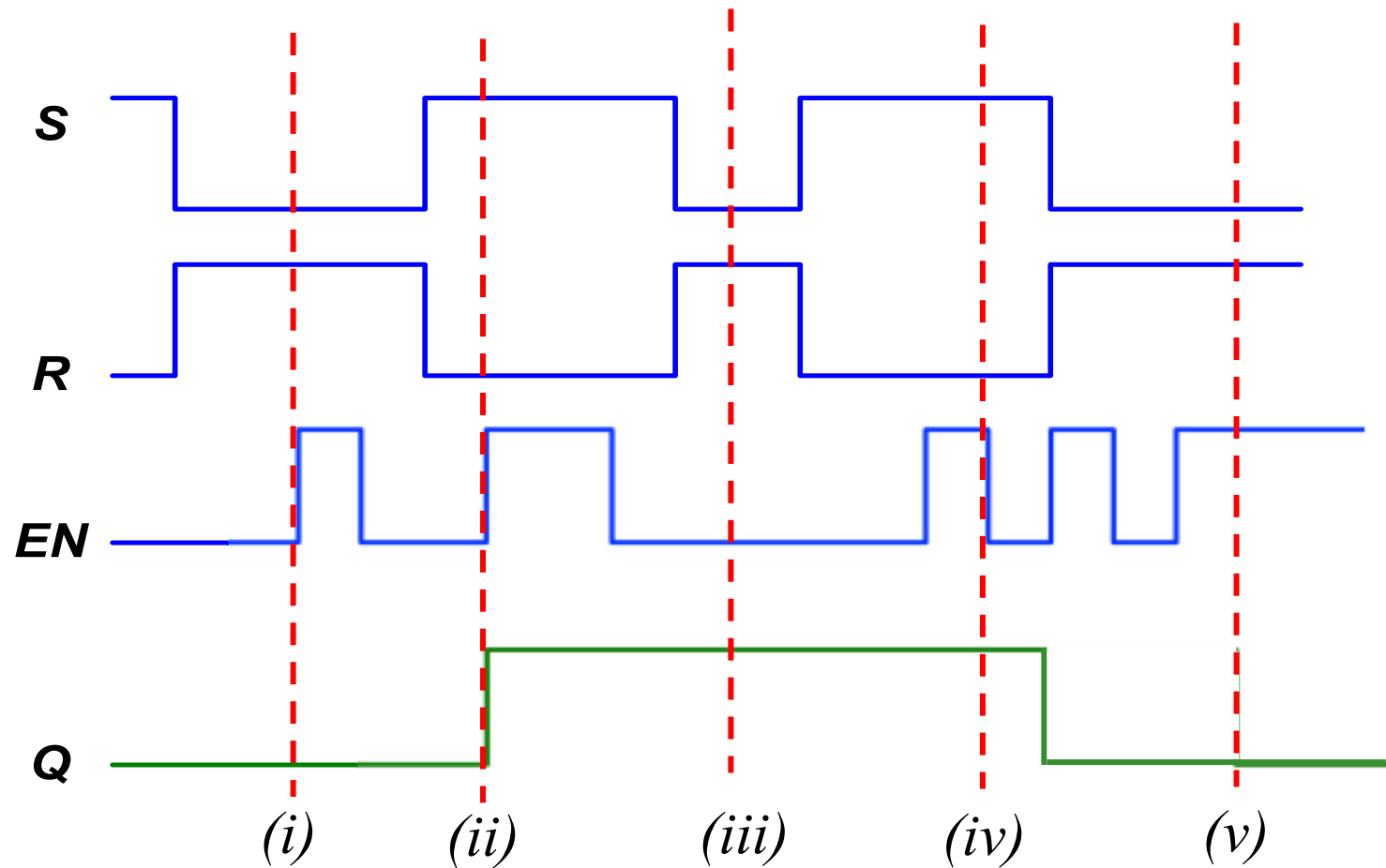
(i) Reset

(ii) Set

(iii) Hold / No change

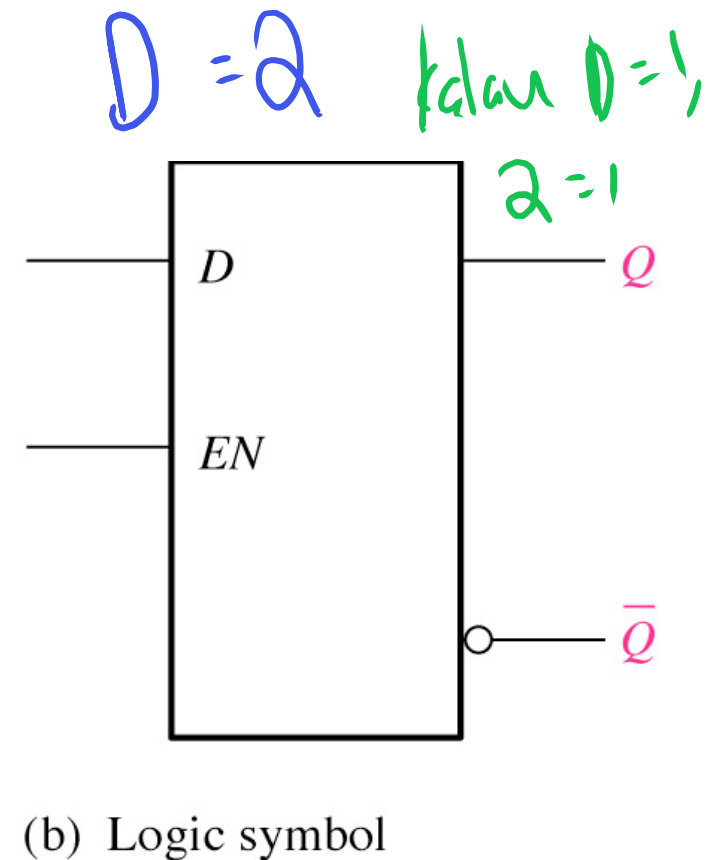
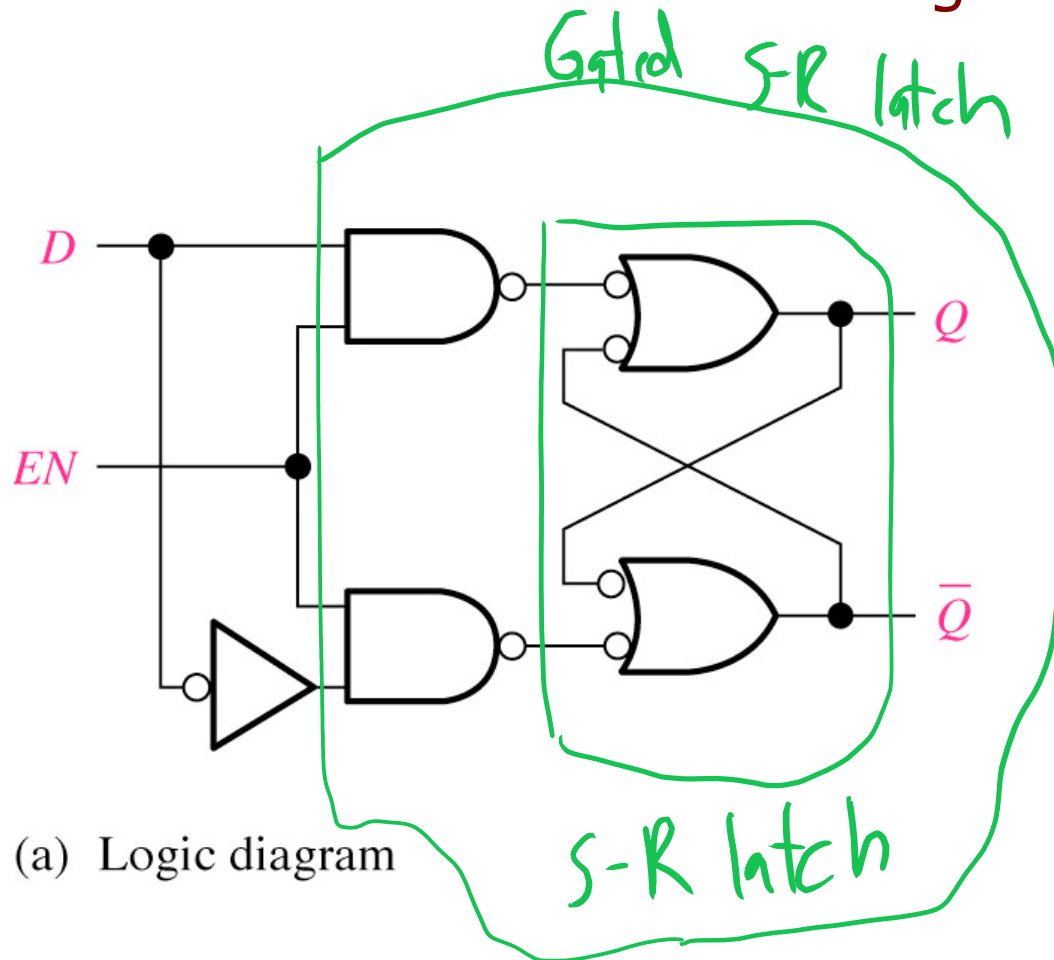
(iv) Hold / No change

(v) Reset

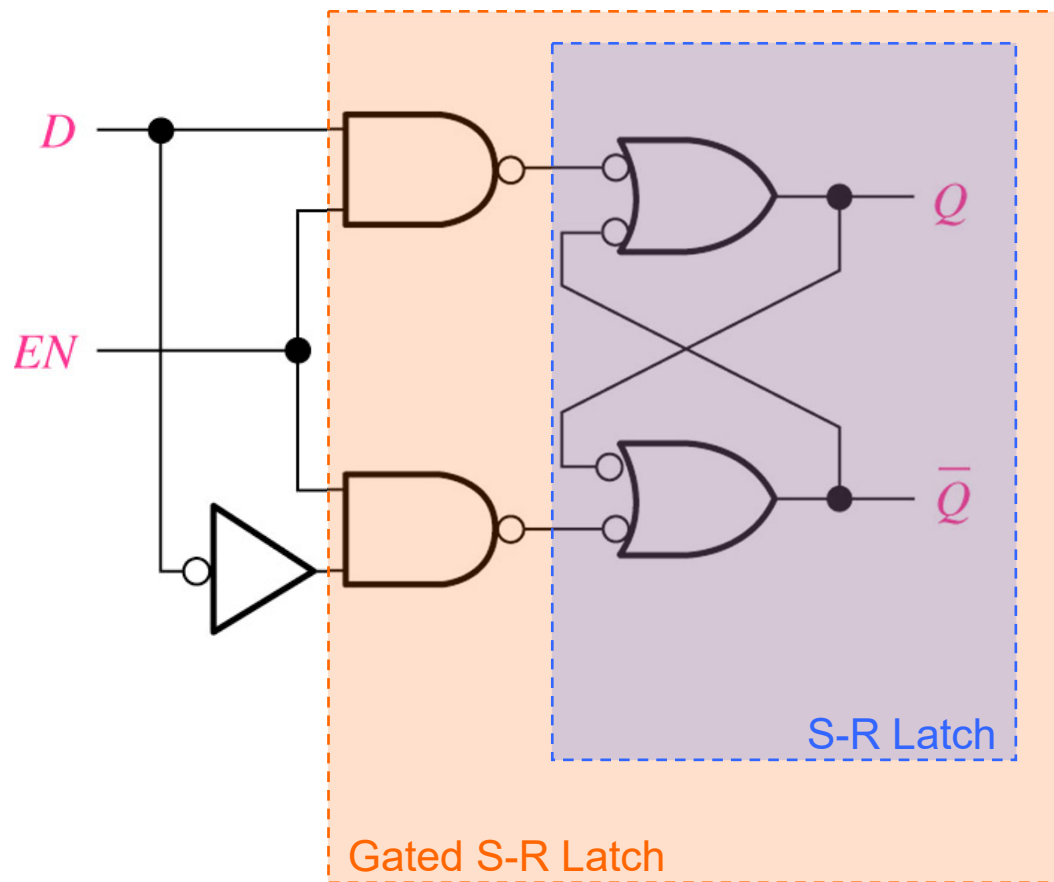


(iii) Gated D Latch:

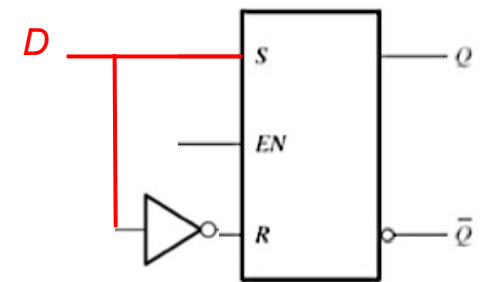
Logic diagram & logic symbol



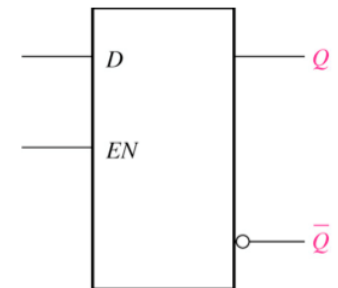
Gated D Latch



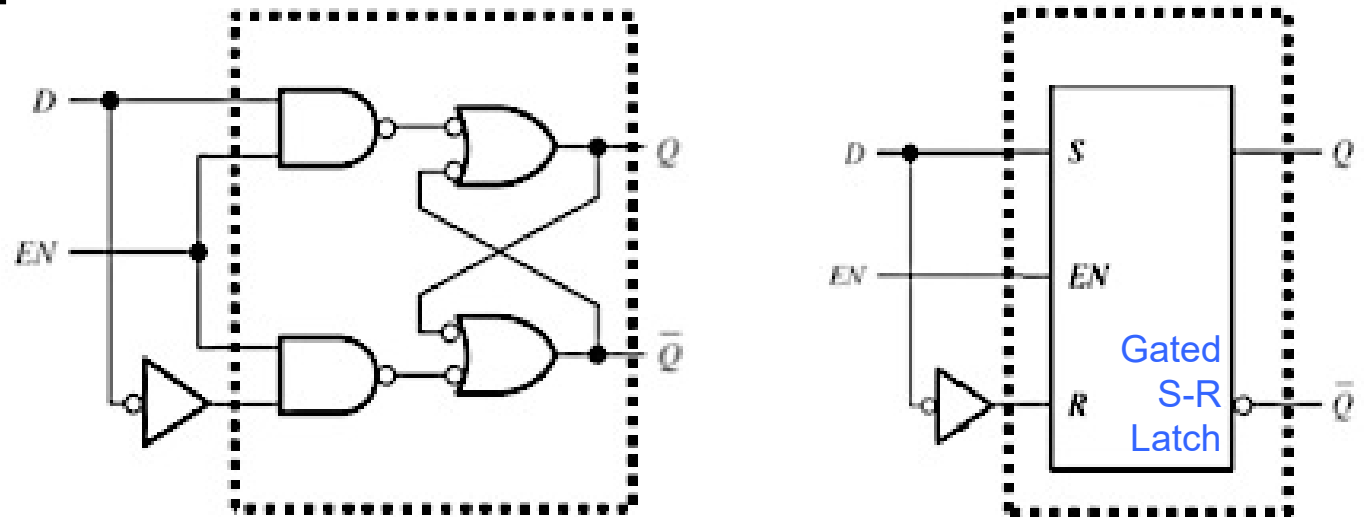
Modified Gated S-R Latch



Gated D Latch



Gated D Latch:



The purpose of the inverter is to make sure that R is the complement of S and R will never be the same as S . So that, we have only two condition at the input of SR.

Condition 1: $D = 0$, therefore $S = 0$, $R = 1$ which make $Q = 0$

Condition 2: $D = 1$, therefore $S = 1$, $R = 0$ which make $Q = 1$

Just like gated SR, when $EN=0$, the output does not change. BUT, if $EN=1$, the output will depend on the value of input D .

So, we can conclude that $Q=D$ when $EN=1$.

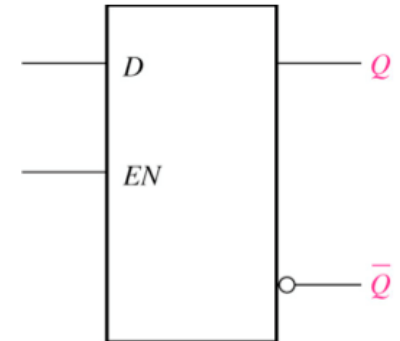
$$EN=1 \rightarrow Q=D$$

Gated D Latch: Truth Table

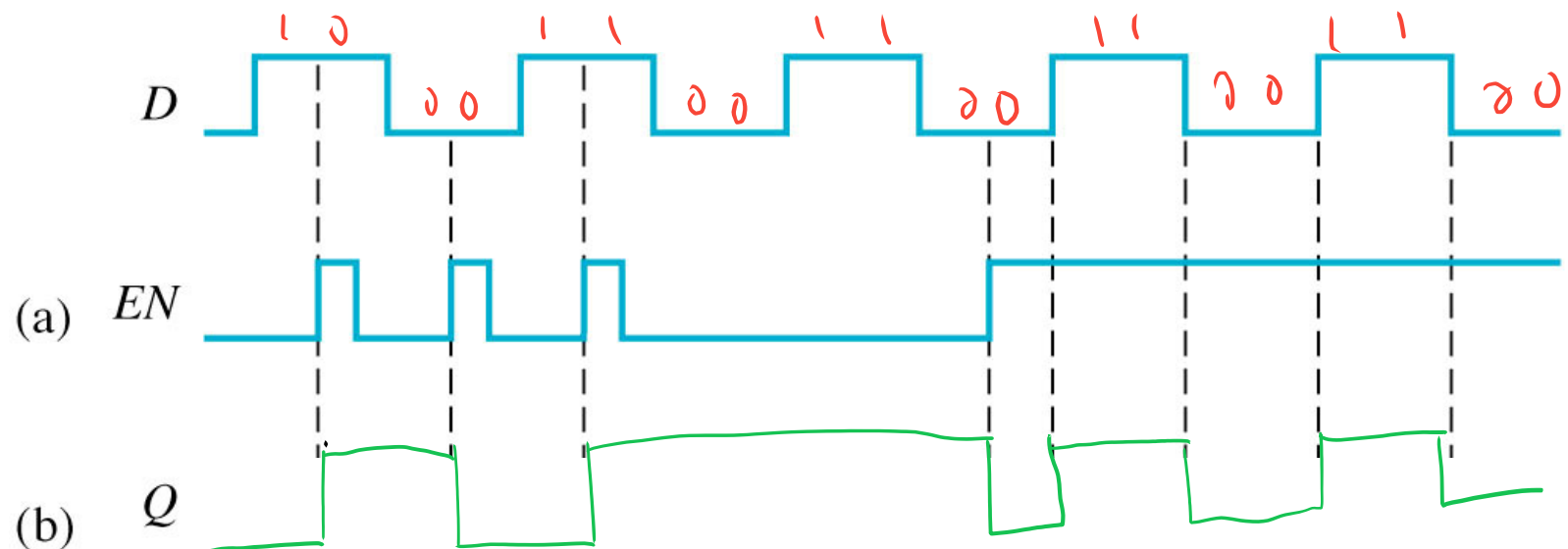
EN	D	Output, Q
0	0	No Change
0	1	No Change
1	0	Q = 0
1	1	Q = 1

EN = 1, Latch is On
EN = 0, Latch is Off

EN = 1 $\rightarrow$ Q = D

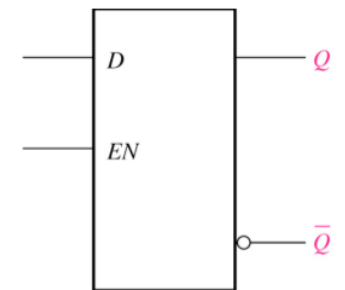


Example: Gated D Latch.
Find the waveform for Q.
Assume that Q is initially LOW.

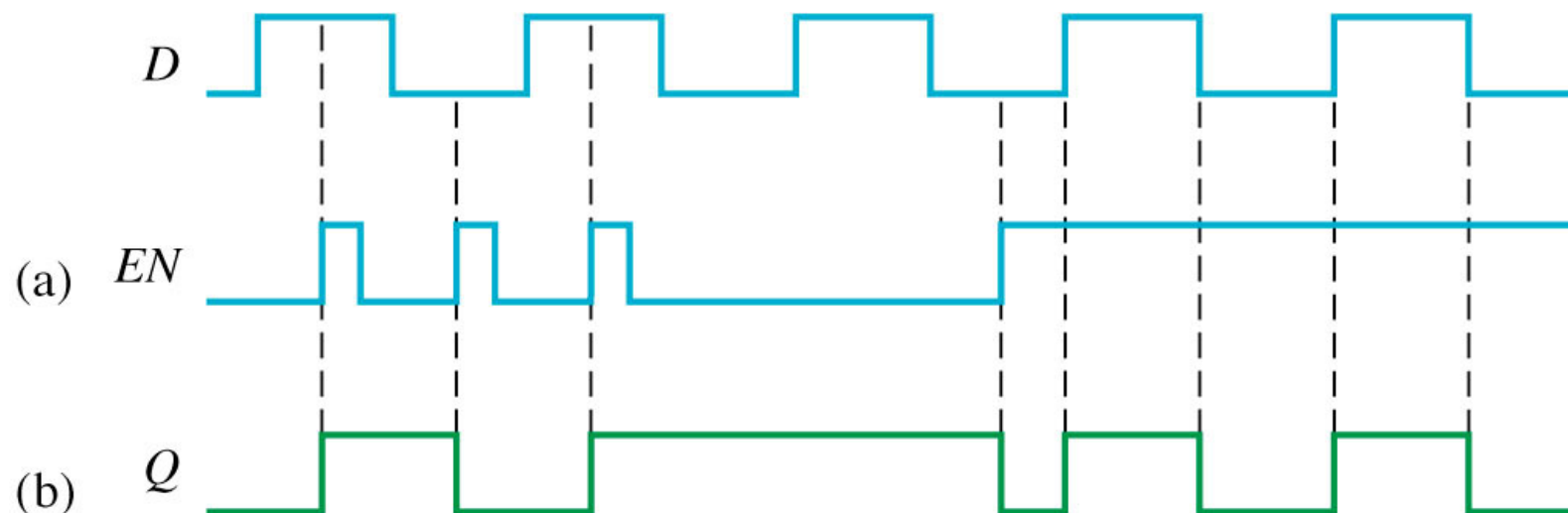


Solution:

EN	D	Output
0	0	No Change
0	1	No Change
1	0	$Q = 0$
1	1	$Q = 1$



Logic symbol





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Flip-Flops



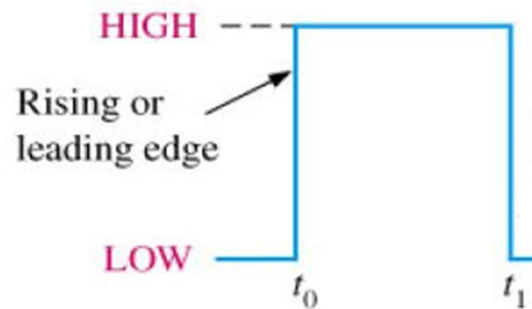
- **Flip-flops (FF)** are **synchronous** bi-stable storage devices capable of storing one bit, where the **output** state only **changes at** a specified point on a **triggering input** called the **Clock (C)**.
- FFs are **edge-triggered**, means that the output changes are synchronized with the **Clock** signal. This may either be
 - a **LOW-to-HIGH** (rising edge) transition or
 - a **HIGH-to-LOW** (falling edge) transition
- Type of Flip-Flops:
 - i) **S-R** iii) **J-K**
 - ii) **D** iv) **T**

Edge-Triggered FF

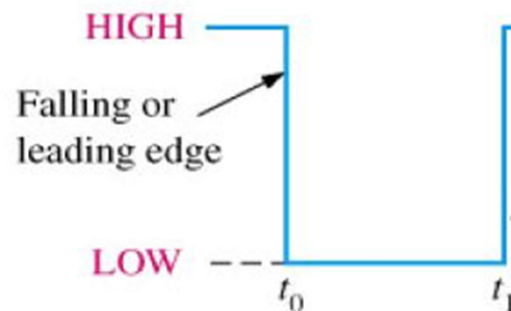
KEY TERMS :

Rising edge = positive edge
Falling edge = negative edge

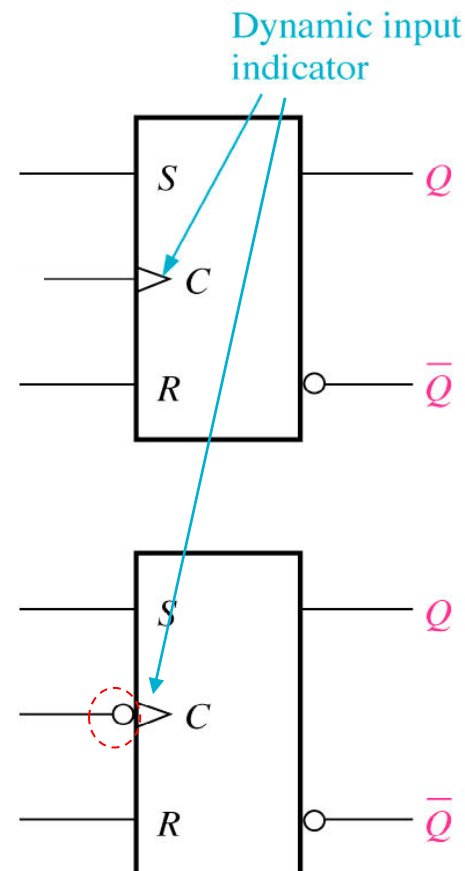
Flip-flop is an important device in a sequential circuit



(a) Positive-going pulse



(b) Negative-going pulse



Edge-Triggered FF logic symbol

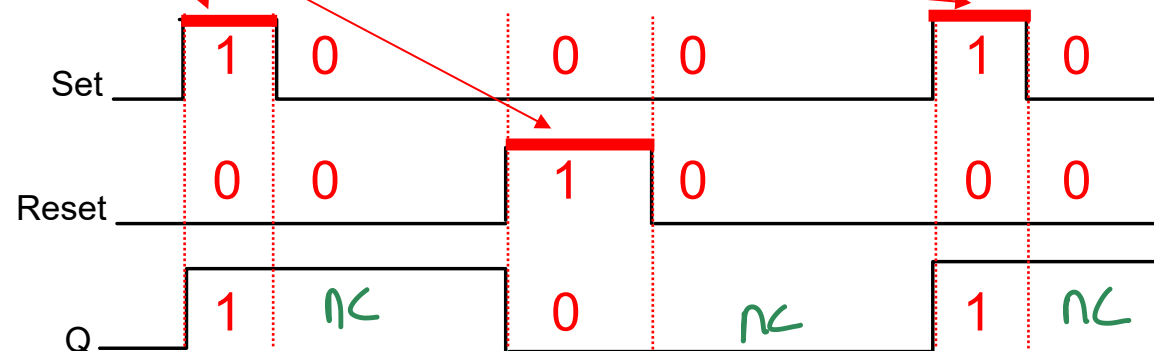
Difference:

Latch vs Flip-Flop

activated by level-triggered,
i.e., HIGH or LOW voltages
from the input signals

Latch

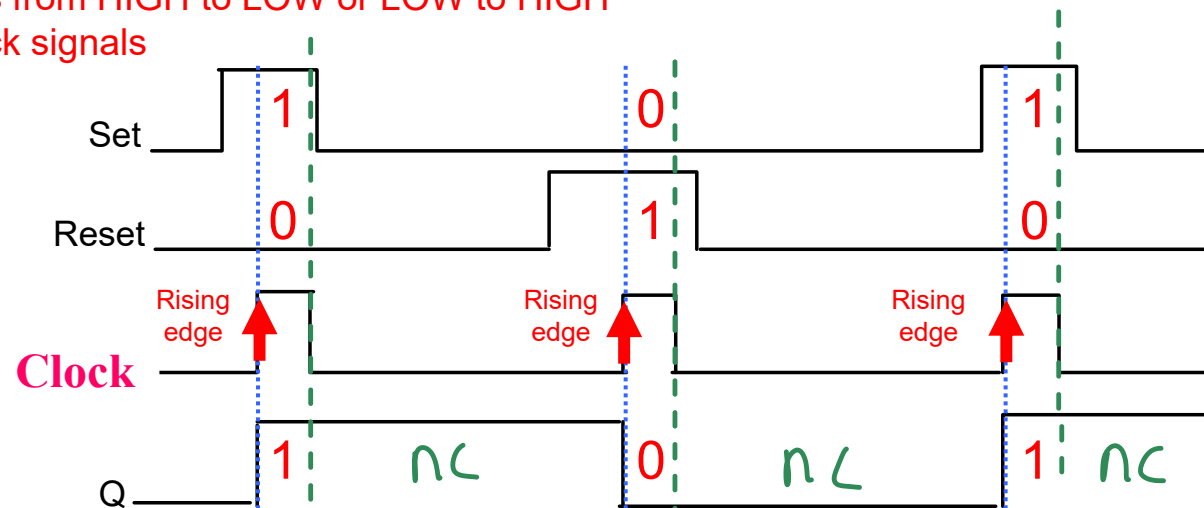
diaktifkan
ikut suis
(S/R)



activated by edge-triggered, i.e., voltage
transitions from HIGH to LOW or LOW to HIGH
at the clock signals

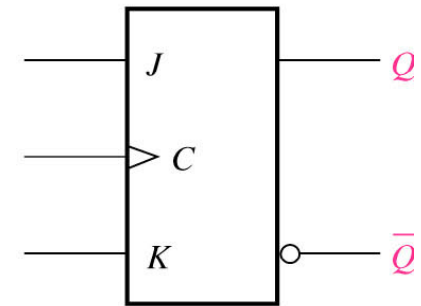
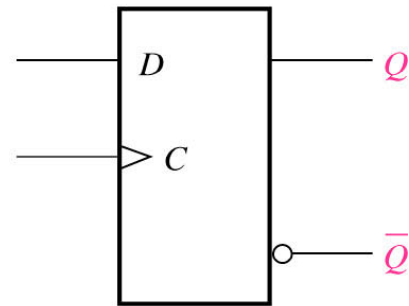
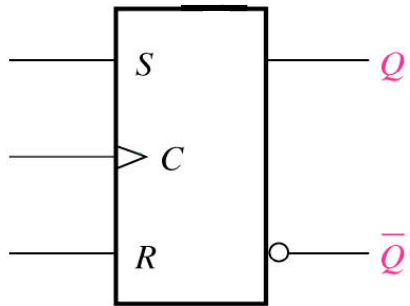
Flip-flop

diaktifkan ikut
clock

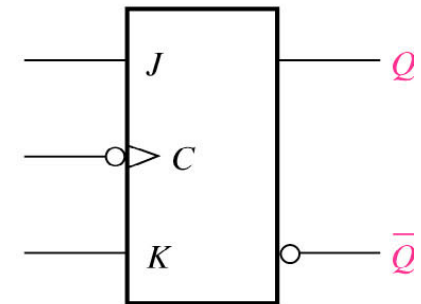
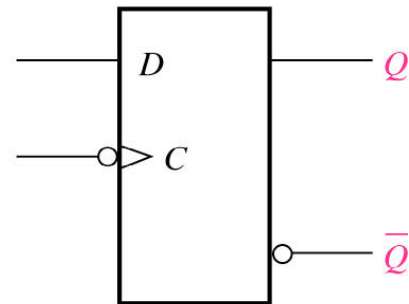
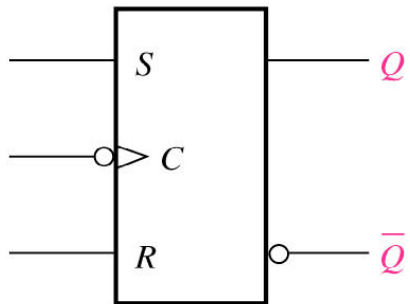


Type of Flip-Flops

(Positive
edge-
triggered)



(Negative
edge-
triggered)

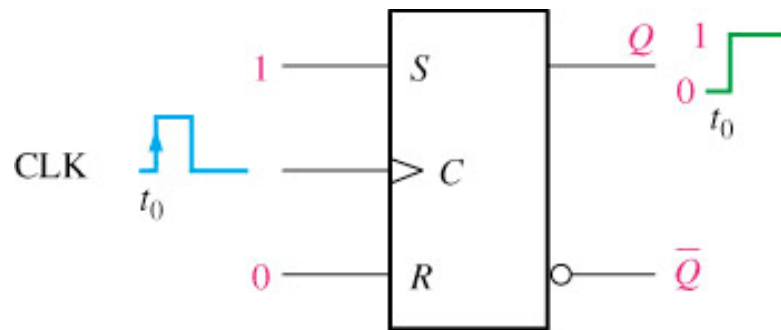


(a) S-R

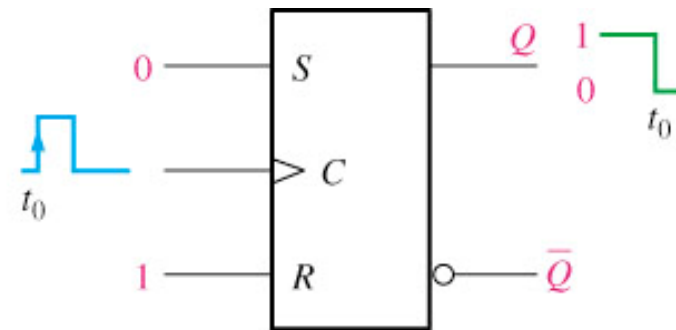
(b) D

(c) J-K

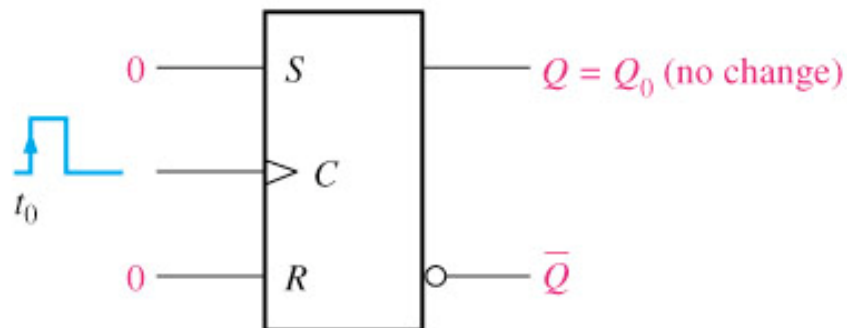
(i) S-R Flip-Flop



(a) $S = 1, R = 0$ flip-flop SETS on positive clock edge. (If already SET, it remains SET.)

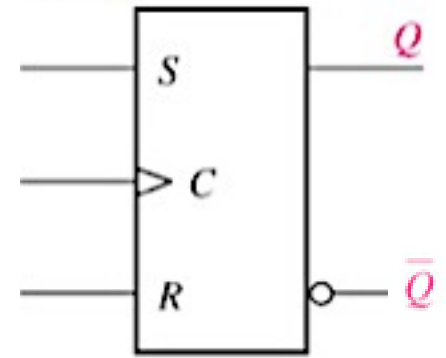


(b) $S = 0, R = 1$ flip-flop RESETS on positive clock edge. (If already RESET, it remains RESET.)



(c) $S = 0, R = 0$ flip-flop does not change. (If SET, it remains SET; if RESET, it remains RESET.)

Konsep hampir sama w/ S-R latch



S	R	CLK	Output		Comments
			Q	$\bar{Q}$	
0	0	↑	Q_0	$\bar{Q}_0$	No Change
0	1	↑	0	1	RESET
1	0	↑	1	0	SET
1	1	↑	?	?	Invalid

TRUTH TABLE

↑ = clock transition LOW to HIGH

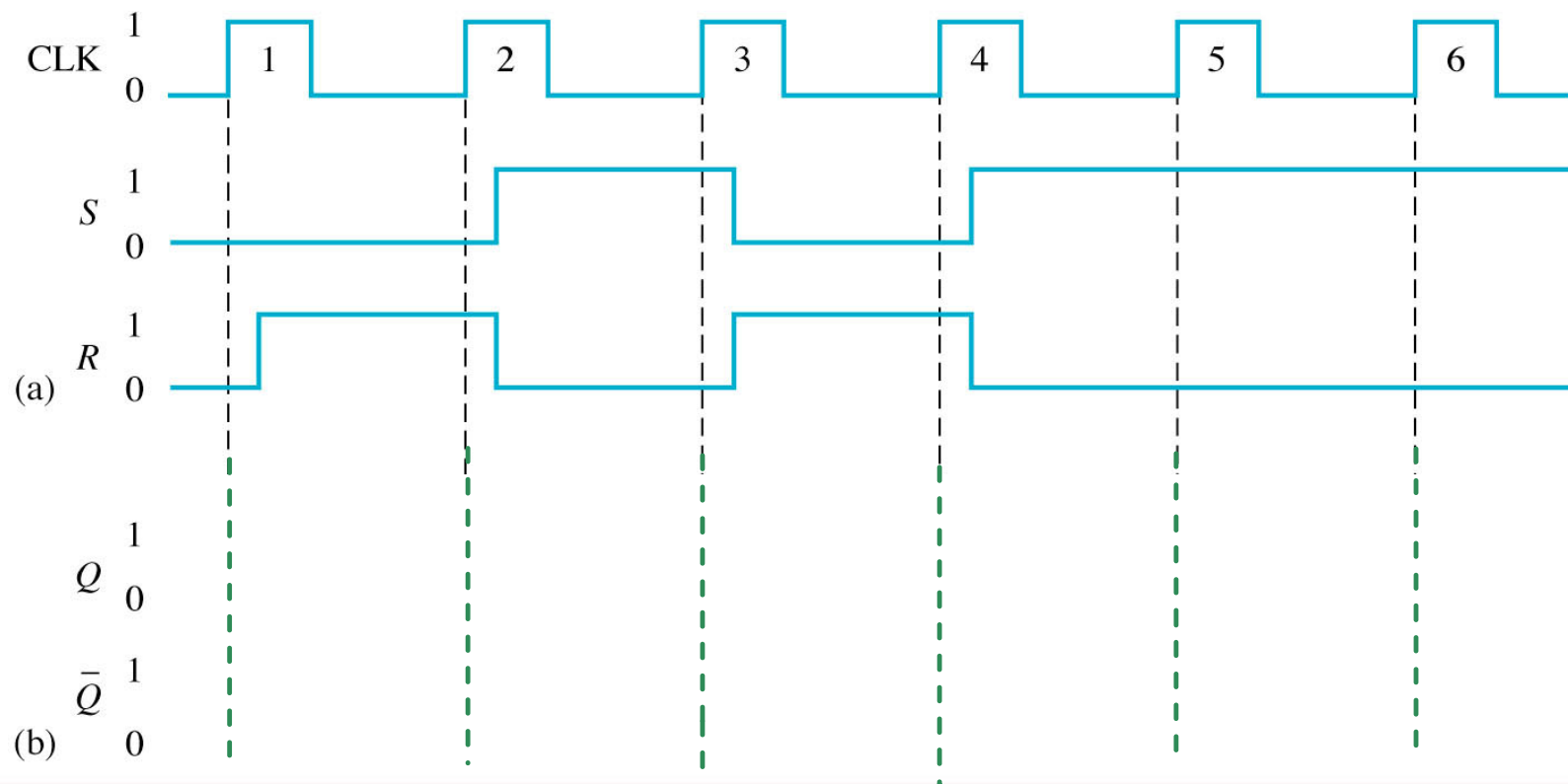
Q_0 = output level prior to clock transition

Example: S-R Flip-Flop.

Jawaban dibawah

Find the waveform for Q and $\bar{Q}$.

Assume that the positive edge-triggered flip-flop is initially RESET.

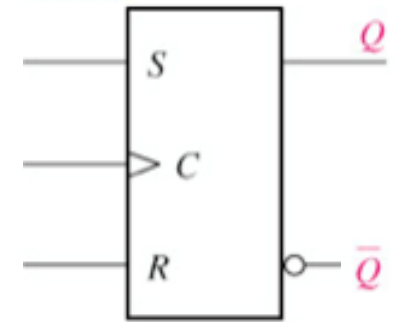


S-R Flip-Flop.

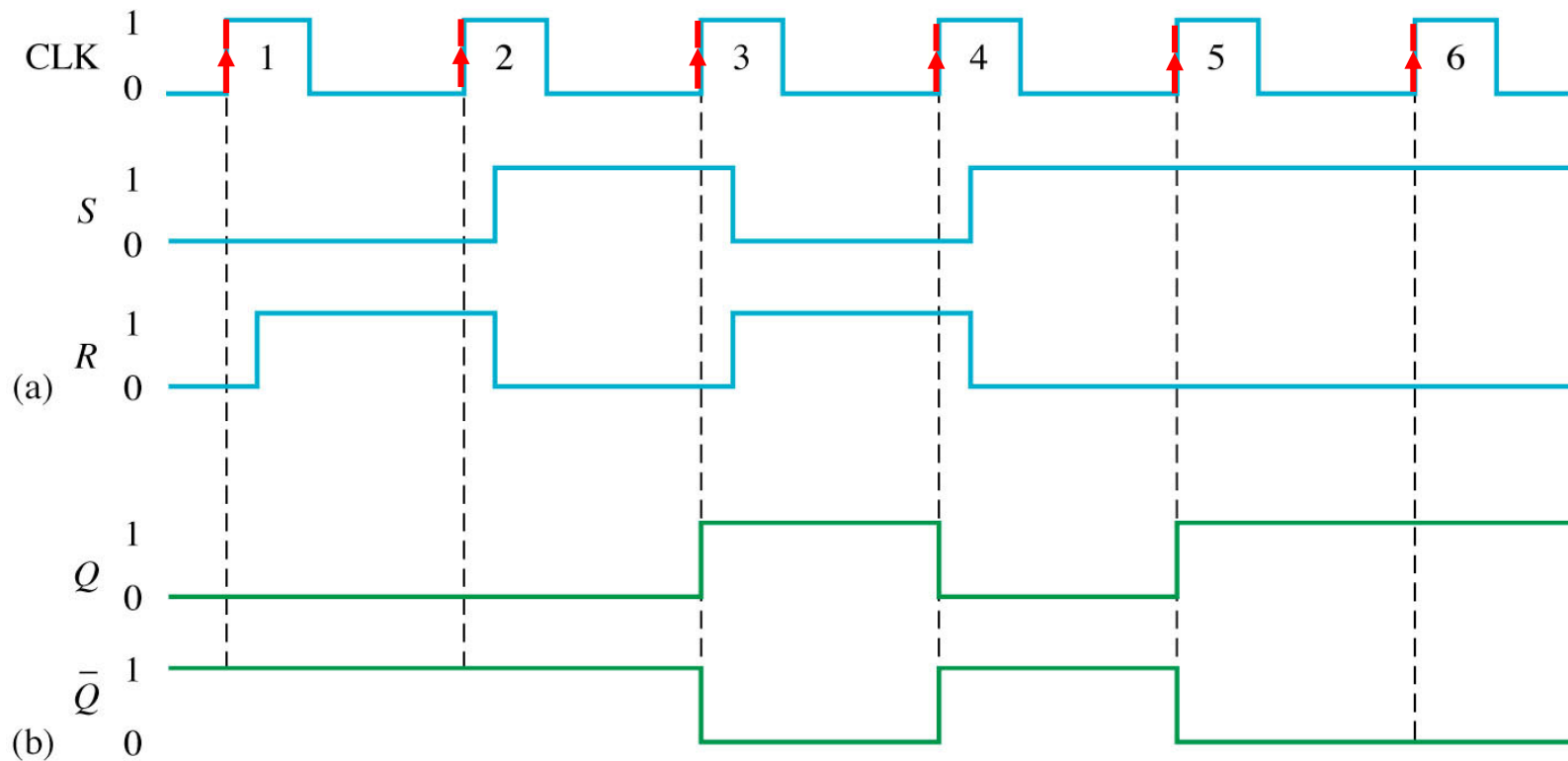
Find the waveform for Q and Q bar.

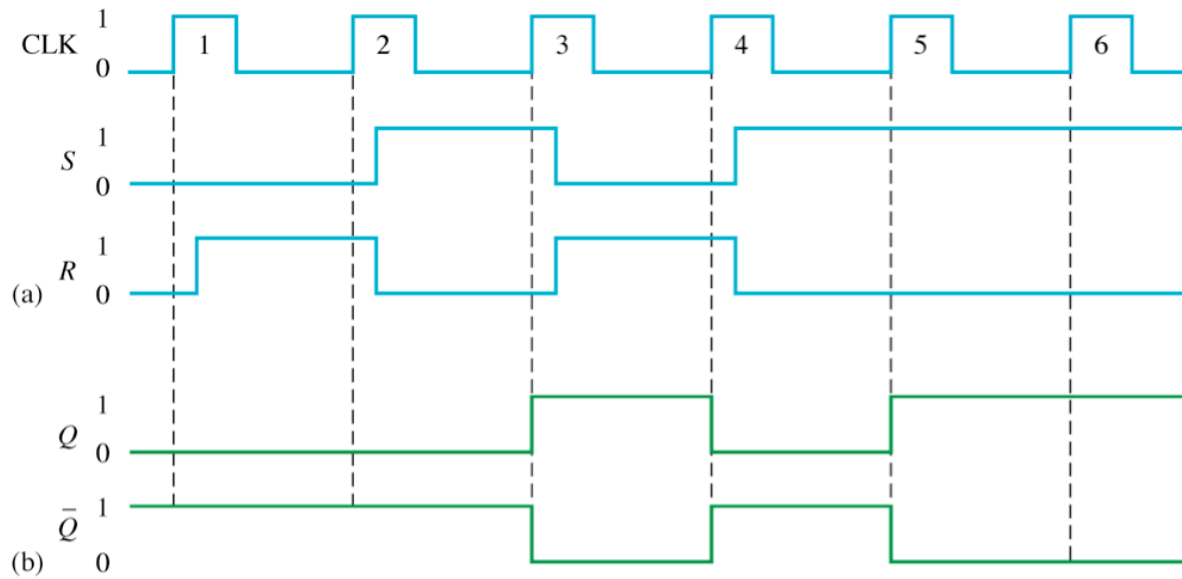
Assume that the flip-flop is initially RESET.

Solution:



Logic symbol



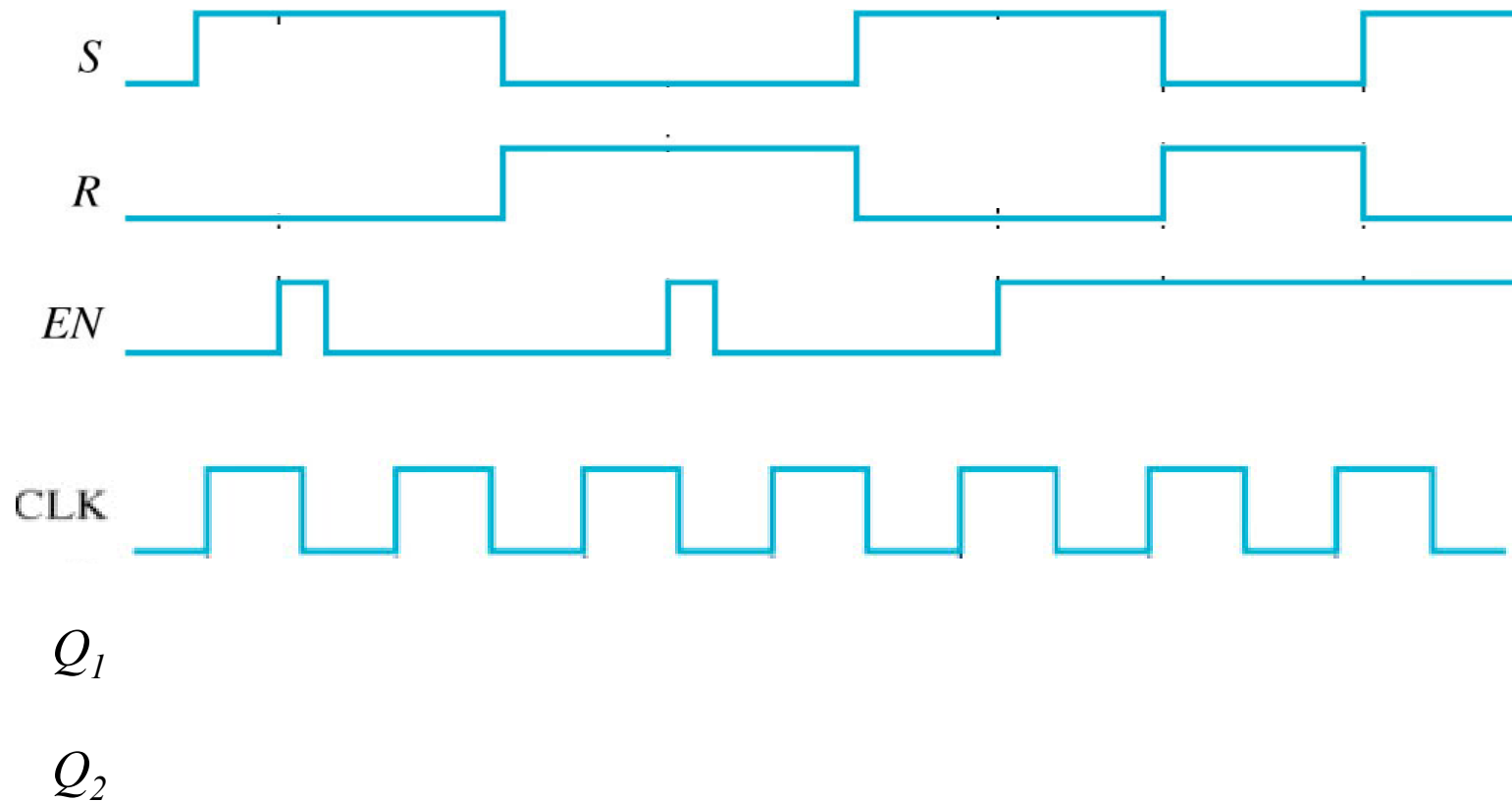


Once Q is determined, $\bar{Q}$ is easily found since it is simply the complement of Q.

Clock Pulse	S	R	Q
1	0	0	Does not change
2	0	1	LOW (RESET)
3	1	0	HIGH (SET)
4	0	1	LOW (RESET)
5	1	0	HIGH (SET)
6	1	0	Stay HIGH

Exercise 7.3:

Given the inputs S and R for Gated S-R Latch with the output Q_1 , and for flip-flop with output Q_2 . Find the waveforms for Q_1 and Q_2 . Assume that both outputs are initially LOW and the flip-flop activated at positive edge-triggered.

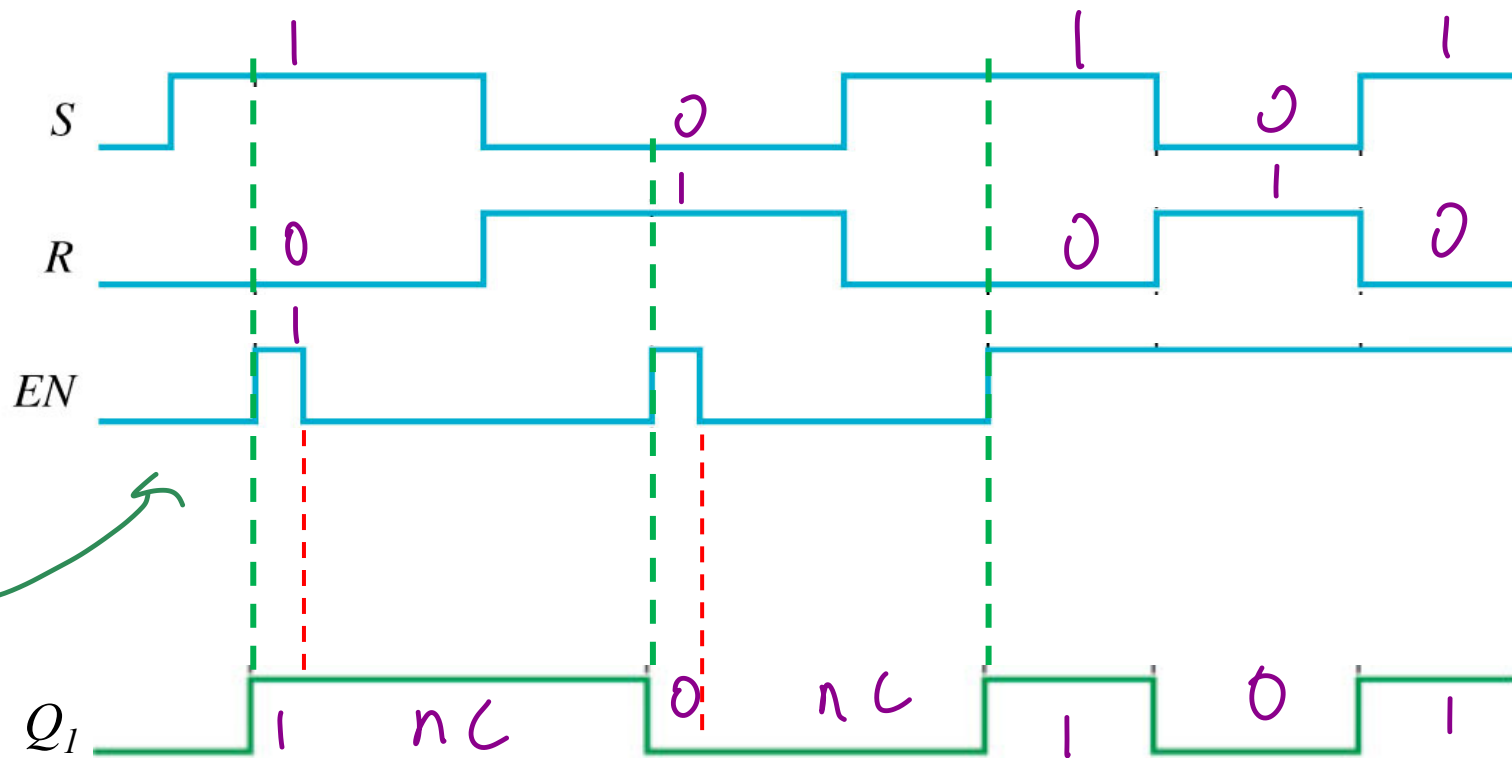


Solution 7.3:

Latch

Gated S-R Latch with the output Q_1

-> waveforms for Q_1 :



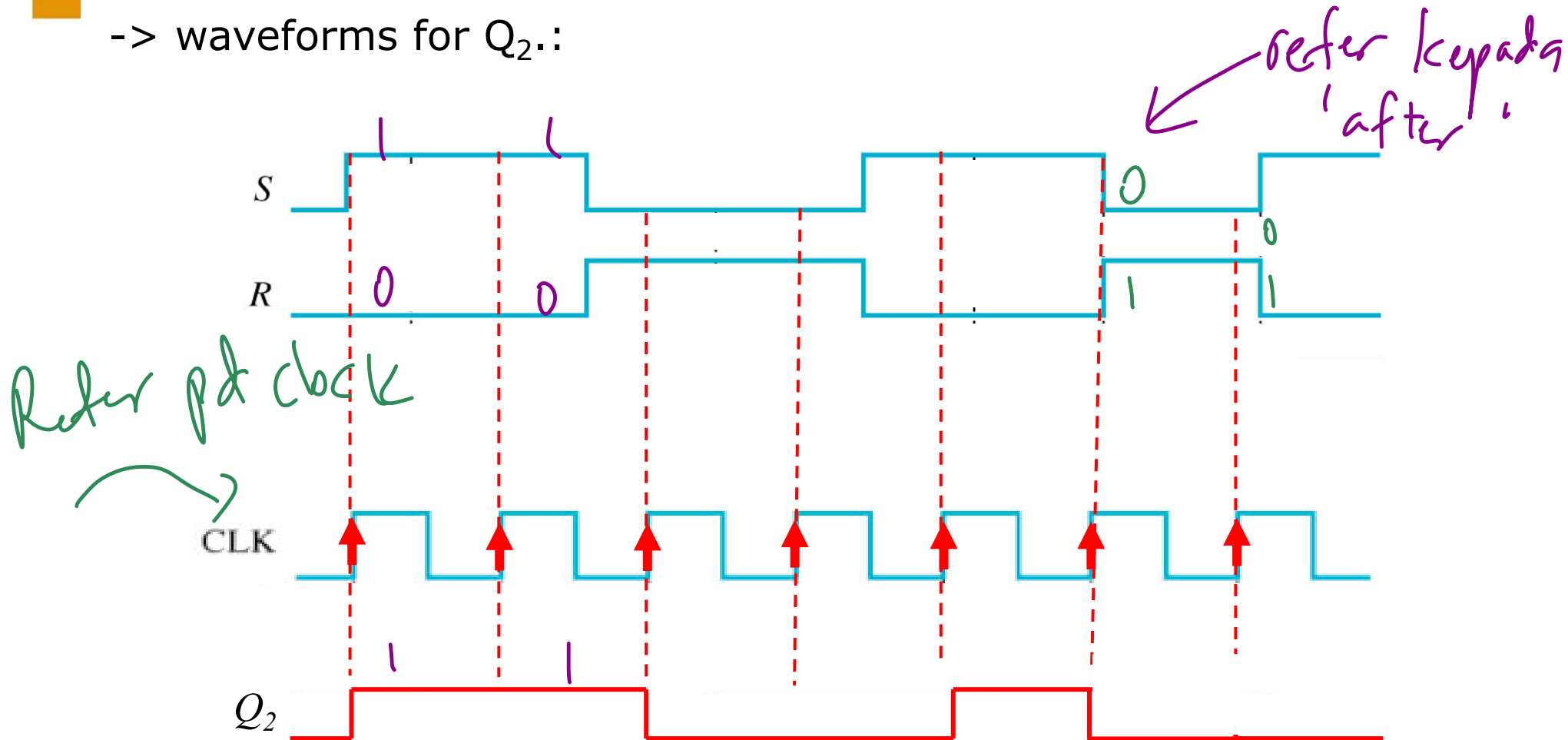
refer
pd EN

Solution 7.3:

Flip-flop

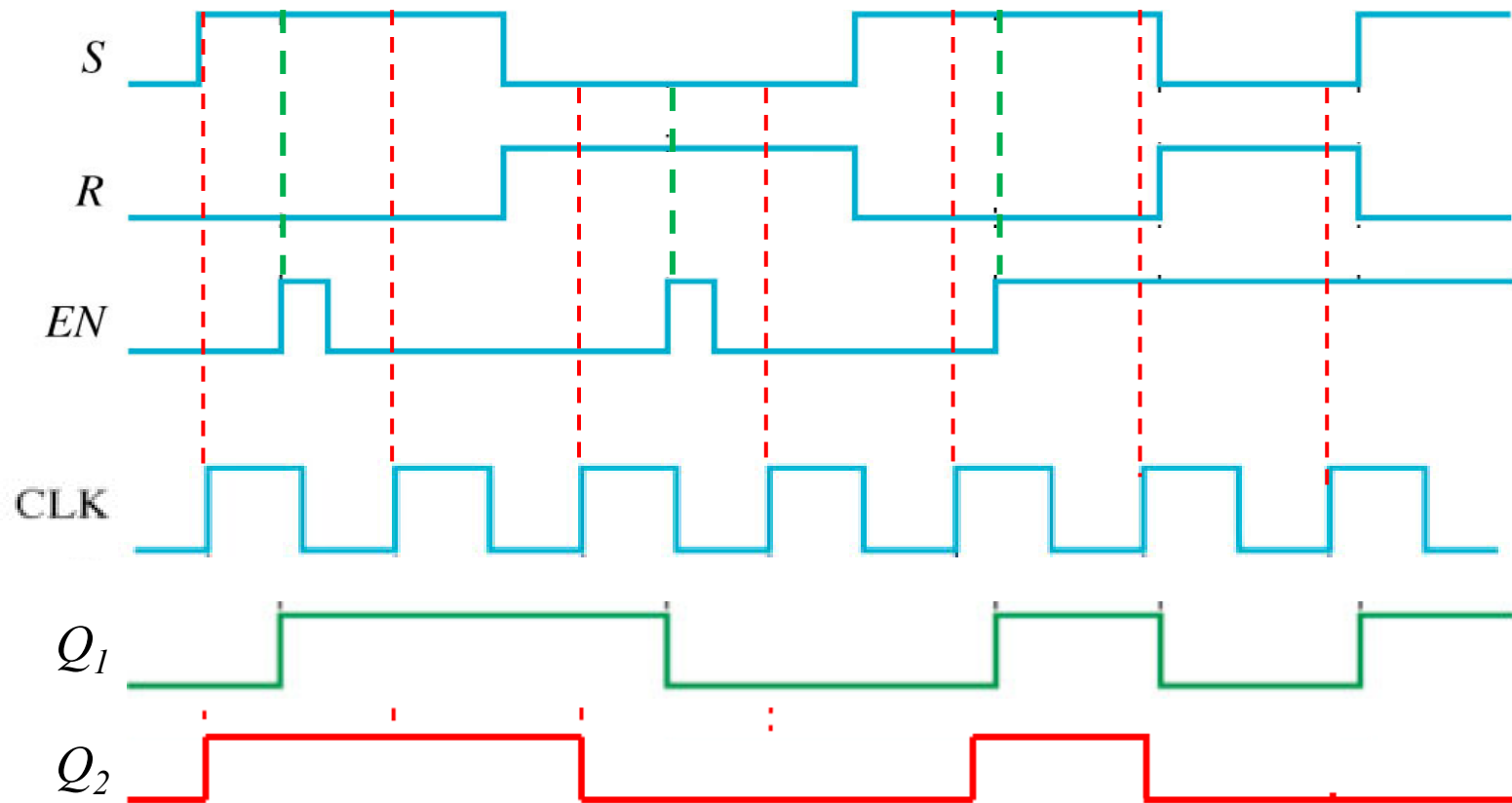
Flip-flop with output Q_2 and activated at **positive edge-triggered**

-> waveforms for Q_2 :



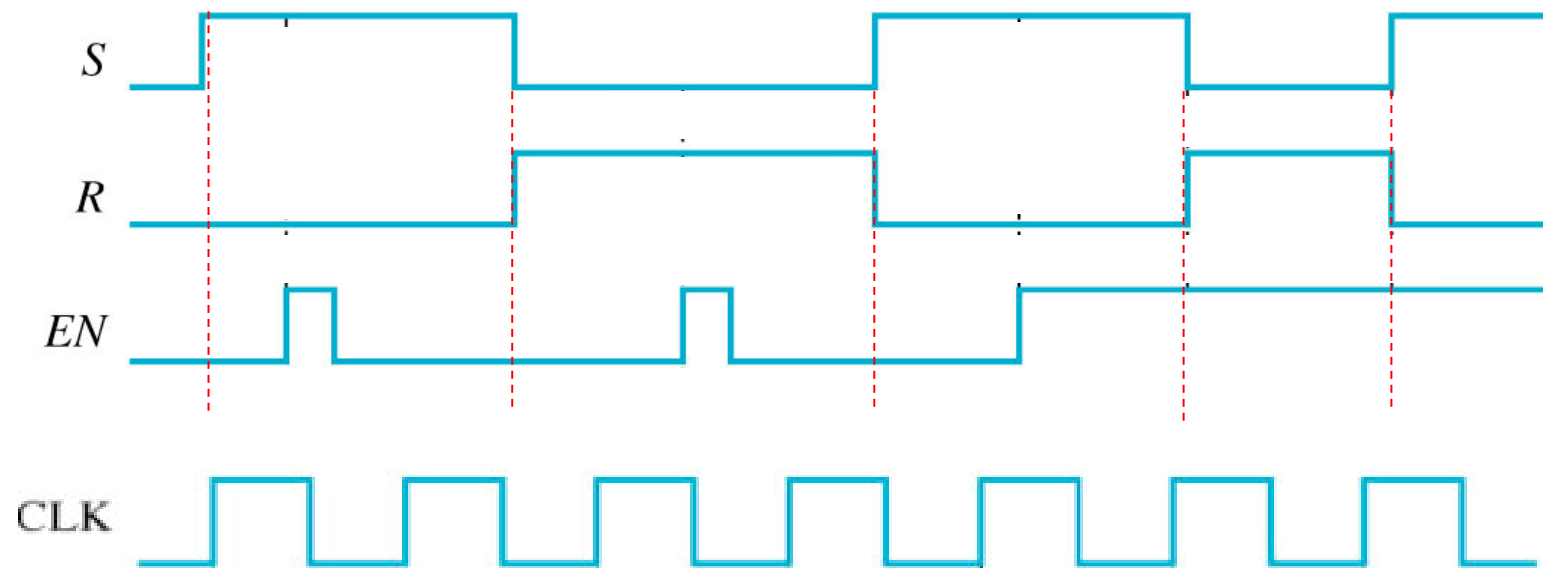
Solution 7.3:

Combined waveforms for Q_1 and Q_2 :



Self-Test:

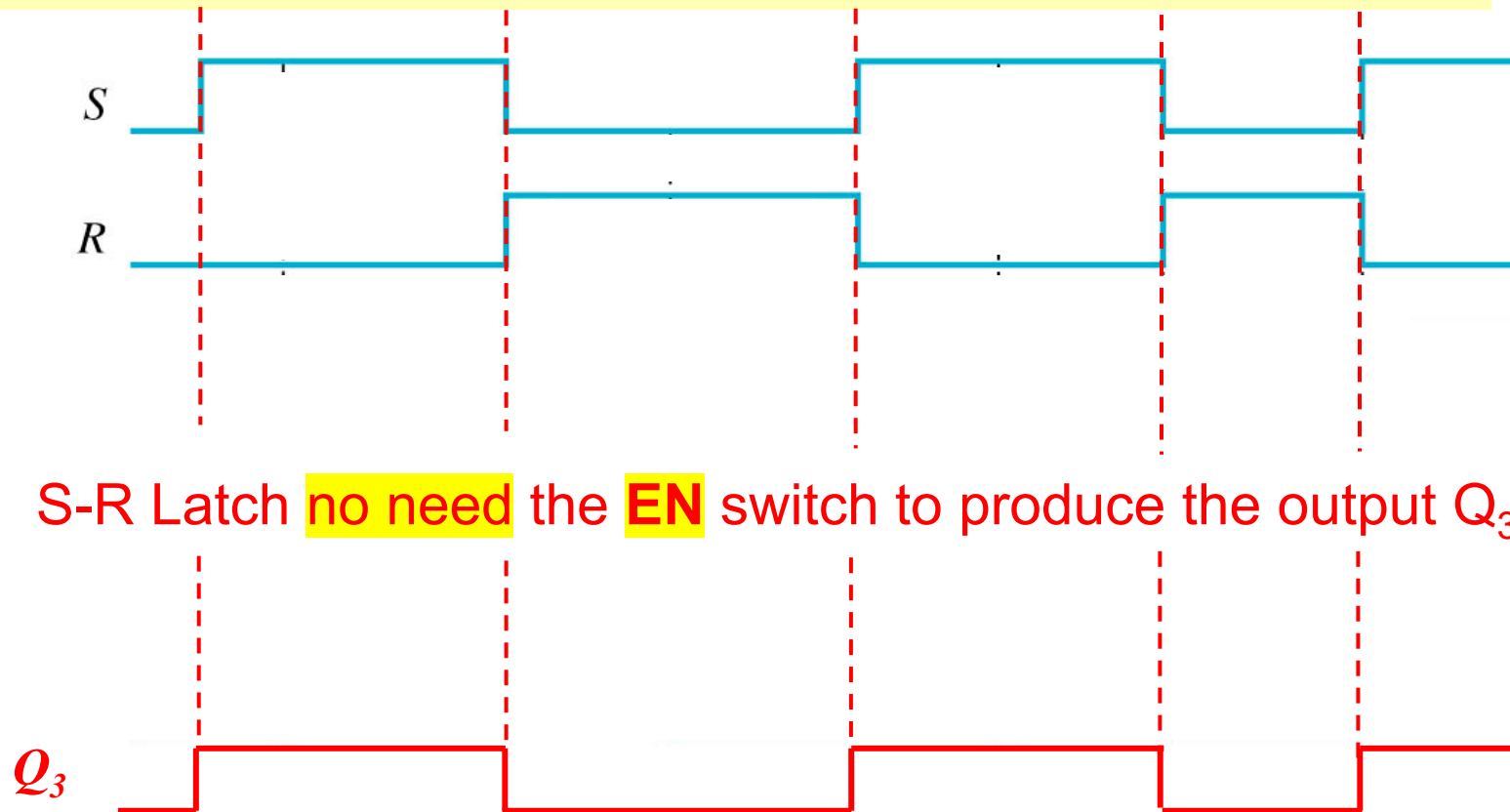
If a S-R Latch is used with output Q_3 , find the waveforms for Q_3 . Is the waveform Q_3 similar to the waveform Q_1 in Exercise 7.3. Justify your answer. Assume that the outputs are initially LOW.



Q_3

Self-Test:

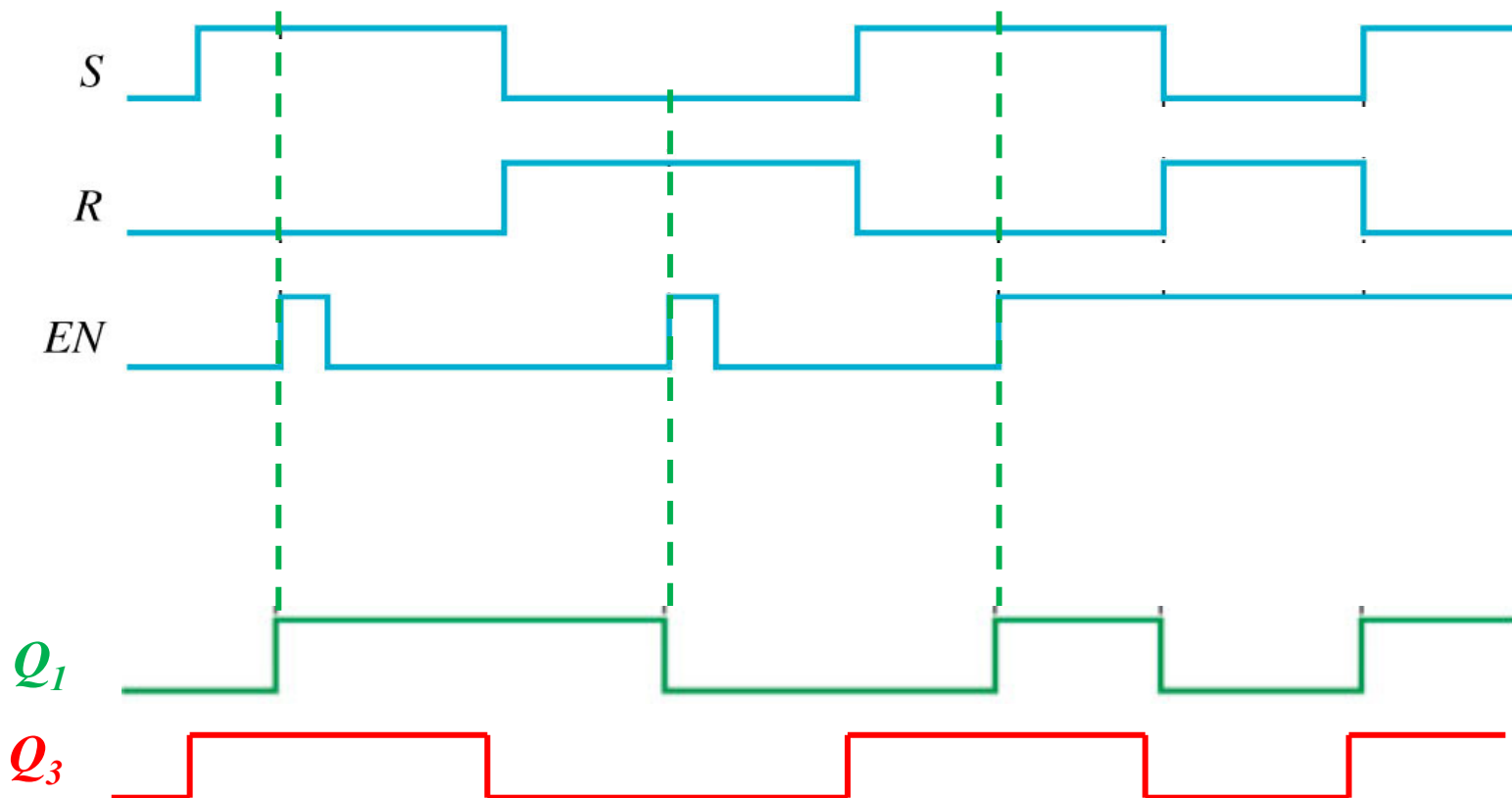
S-R Latch is used with output Q_3 , ... the waveforms for Q_3 .



S-R Latch no need the **EN** switch to produce the output Q_3

Self-Test:

Is the waveforms Q_3 similar to the waveform Q_1 in Exercise 7.3 ? Justify your answer.

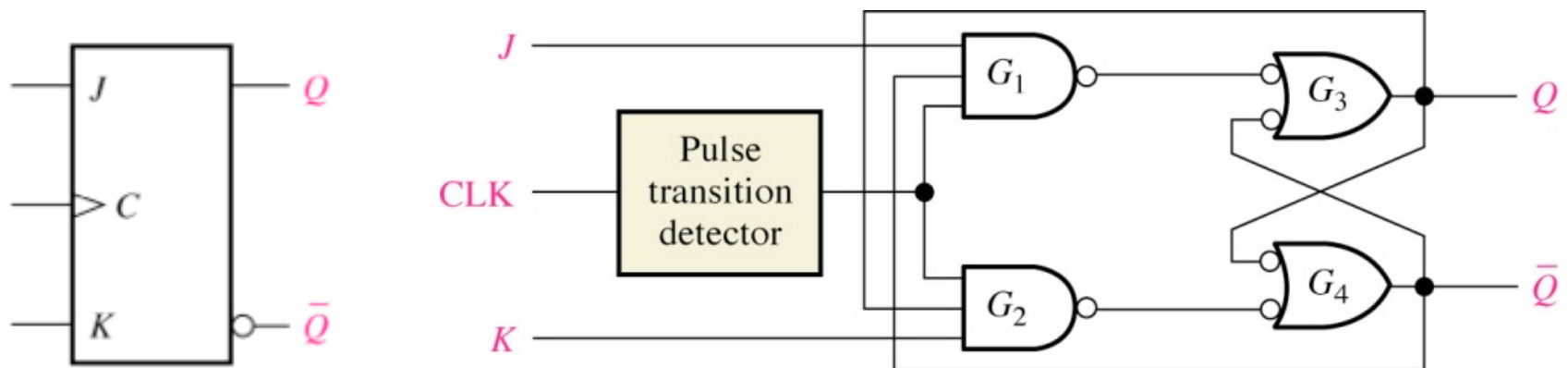


(ii) J-K Flip-Flop

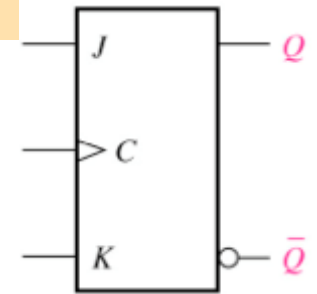
→ universal

- The JK FF is versatile and is a widely used type of FF.
- The difference between J-K and S-R:

J-K has no invalid state as S-R.



Logic Symbol



Logic symbol

J	K	CLK	Output		Comments
			Q	$\bar{Q}$	
0	0	$\uparrow/\downarrow$	Q_0	$\bar{Q}_0$	No Change
0	1	$\uparrow/\downarrow$	0	1	RESET
1	0	$\uparrow/\downarrow$	1	0	SET
1	1	$\uparrow/\downarrow$	$\bar{Q}_0$	Q_0	Toggle

TRUTH TABLE

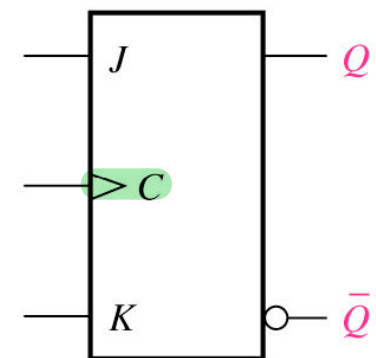
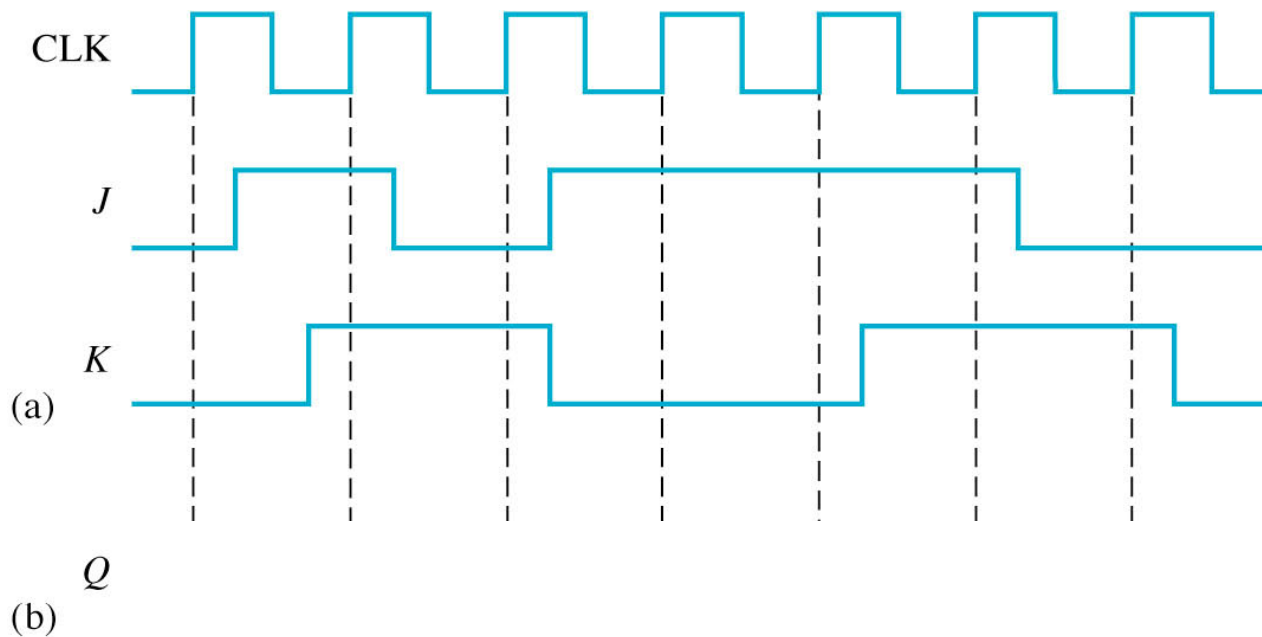
$\uparrow$ = clock transition LOW to HIGH

Q_0 = output level prior to clock transition

Example: J-K Flip-Flop.
Find the waveform for Q.
Assume that Q is initially LOW.

Positive
or
negative
triggered?

positive
→ triggered



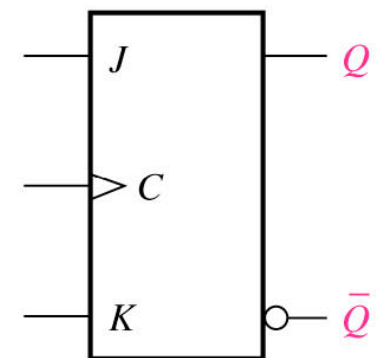
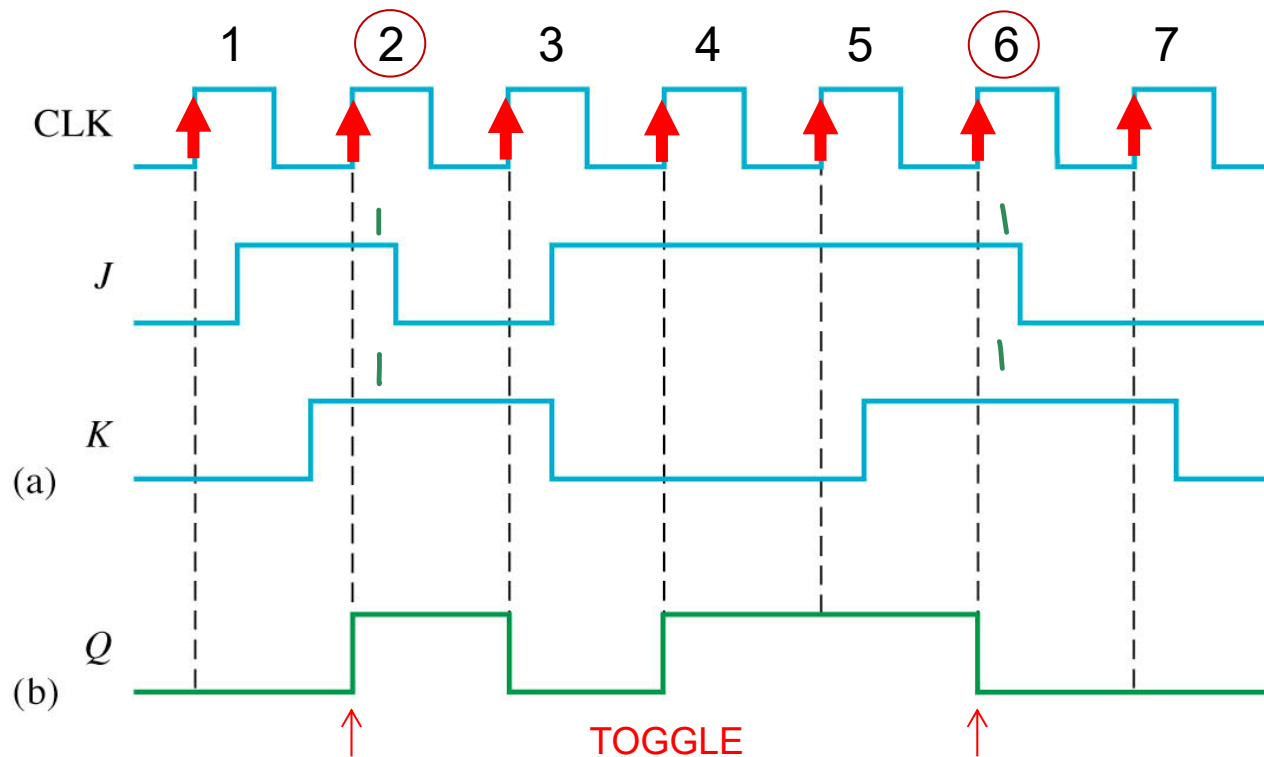
Logic symbol

Solution:

J	K	CLK	Output		Comments
			Q	$\bar{Q}$	
0	0	$\uparrow/\downarrow$	Q_0	$\bar{Q}_0$	No Change
0	1	$\uparrow/\downarrow$	0	1	RESET
1	0	$\uparrow/\downarrow$	1	0	SET
1	1	$\uparrow/\downarrow$	$\bar{Q}_0$	Q_0	Toggle

Positive triggered

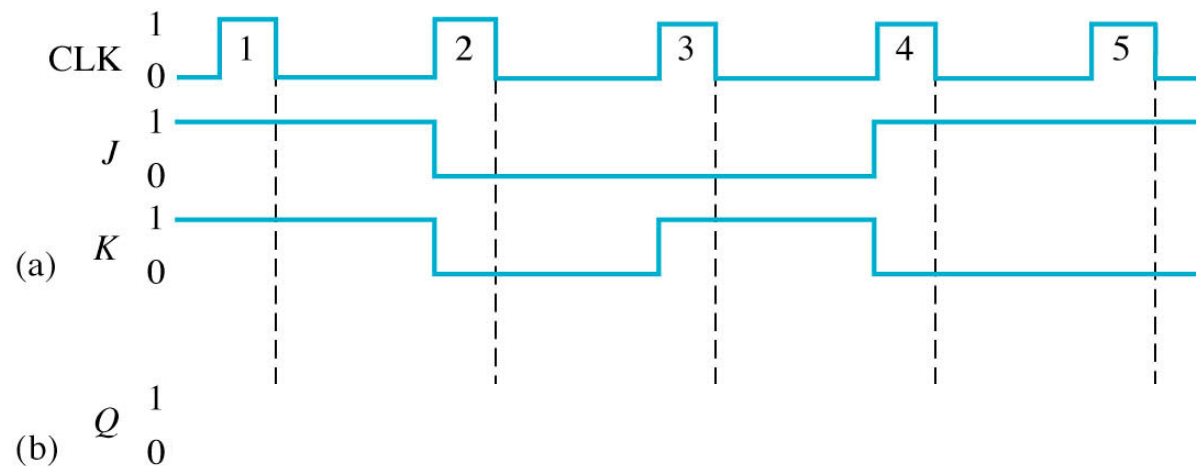
When the output toggled?



Logic symbol

Example: J-K Flip-Flop.
Find the waveform for Q.
Assume that Q is initially LOW.

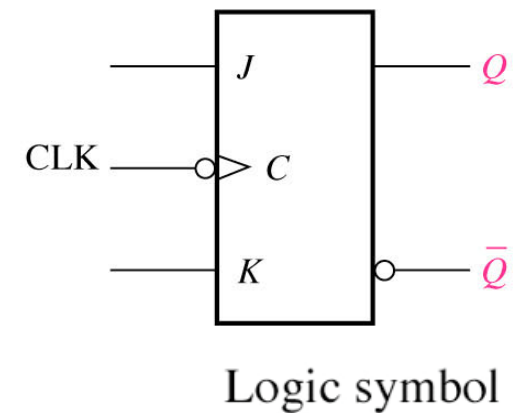
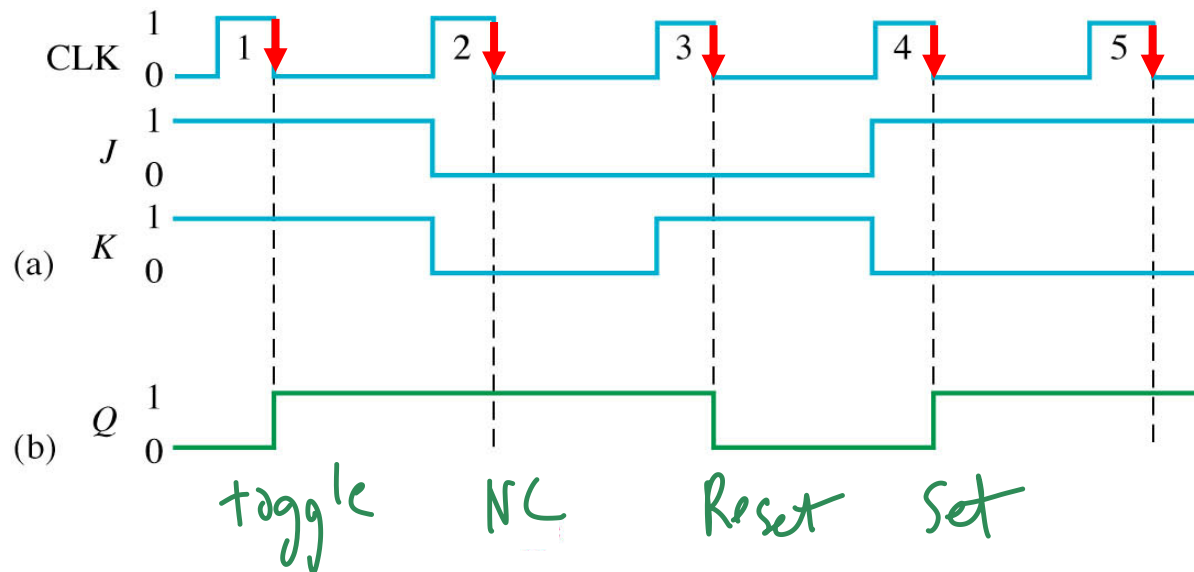
Positive
or
negative
triggered?



Solution:

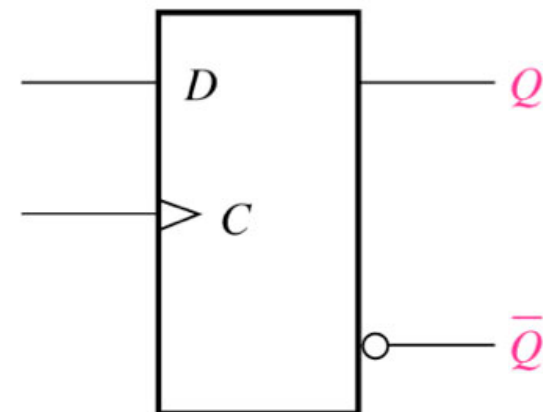
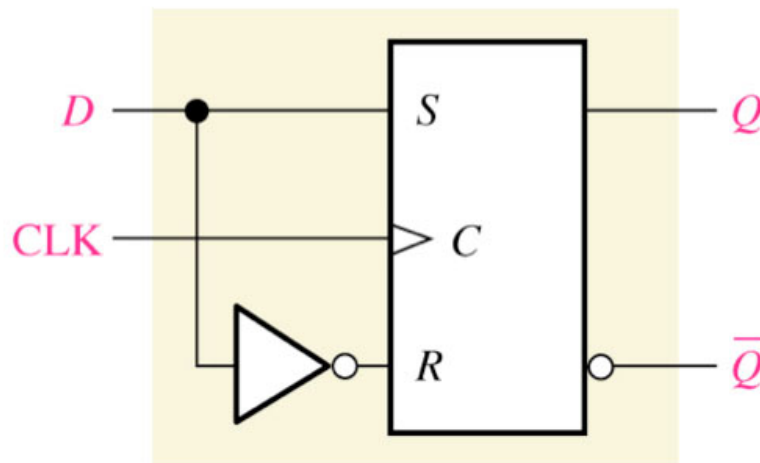
J	K	CLK	Output		Comments
			Q	$\bar{Q}$	
0	0	$\uparrow/\downarrow$	Q_0	$\bar{Q}_0$	No Change
0	1	$\uparrow/\downarrow$	0	1	RESET
1	0	$\uparrow/\downarrow$	1	0	SET
1	1	$\uparrow/\downarrow$	$\bar{Q}_0$	Q_0	Toggle

negative triggered



D → Data or Delay

- D FF is useful when a single data bit (1 or 0) is to be stored.
- The addition of an **inverter** to an S-R FF creates basic D FF where a **positive edge-triggered** type is shown below:



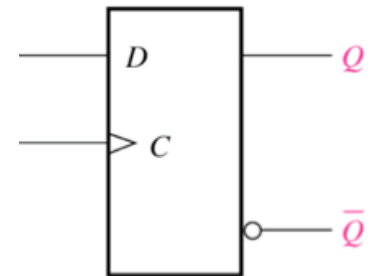
Logic symbol

TRUTH TABLE

INPUTS		OUTPUTS		COMMENTS
D	CLK	Q	$\bar{Q}$	
1	↑	1	0	SET (store a 1)
0	↑	0	1	RESET (store 0)

Q follow D

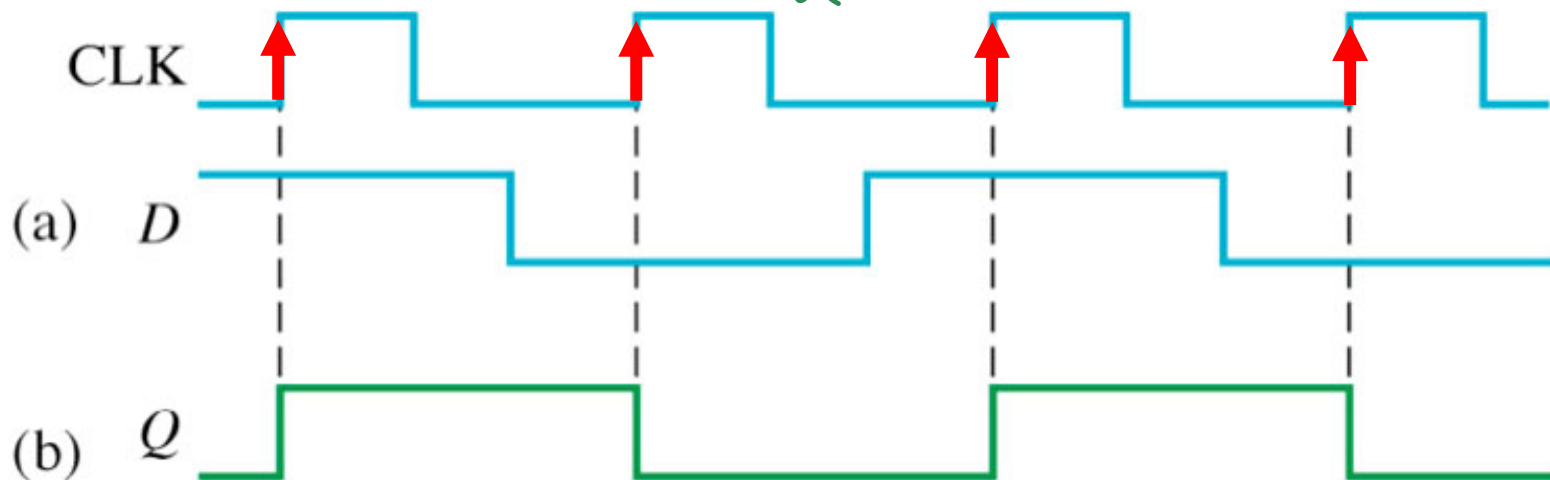
*D = 1,
Q = 1*



Logic symbol

(for positive edge-triggered)

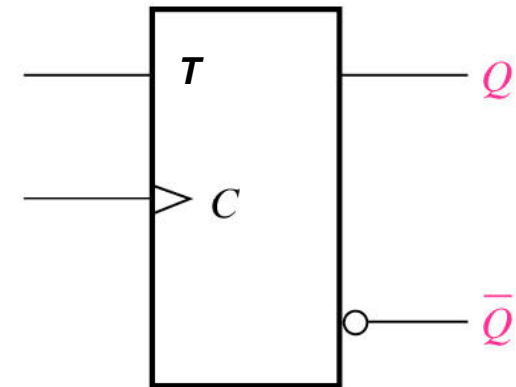
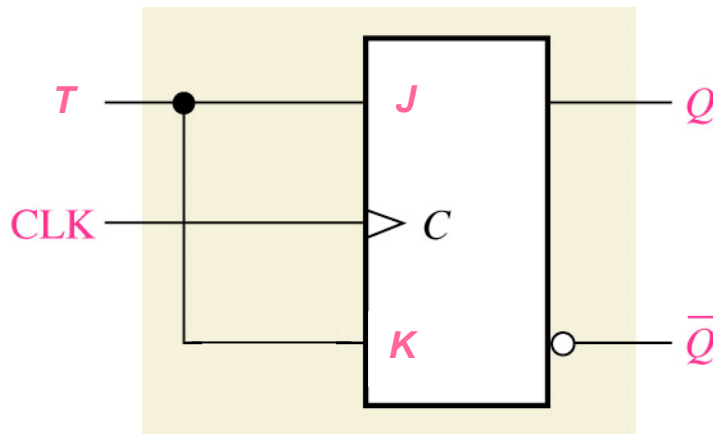
*D = 0,
Q = 0*



Remember, Q follows D at the active triggering clock edge.

(iv) T Flip-Flop

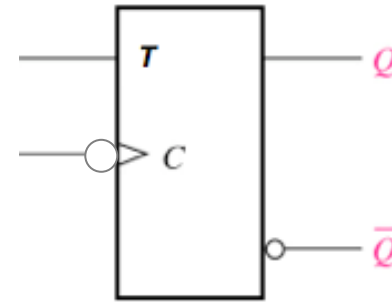
- Also called as **toggle** flip-flop.
- Frequently used in building **counters**.



Logic Symbol

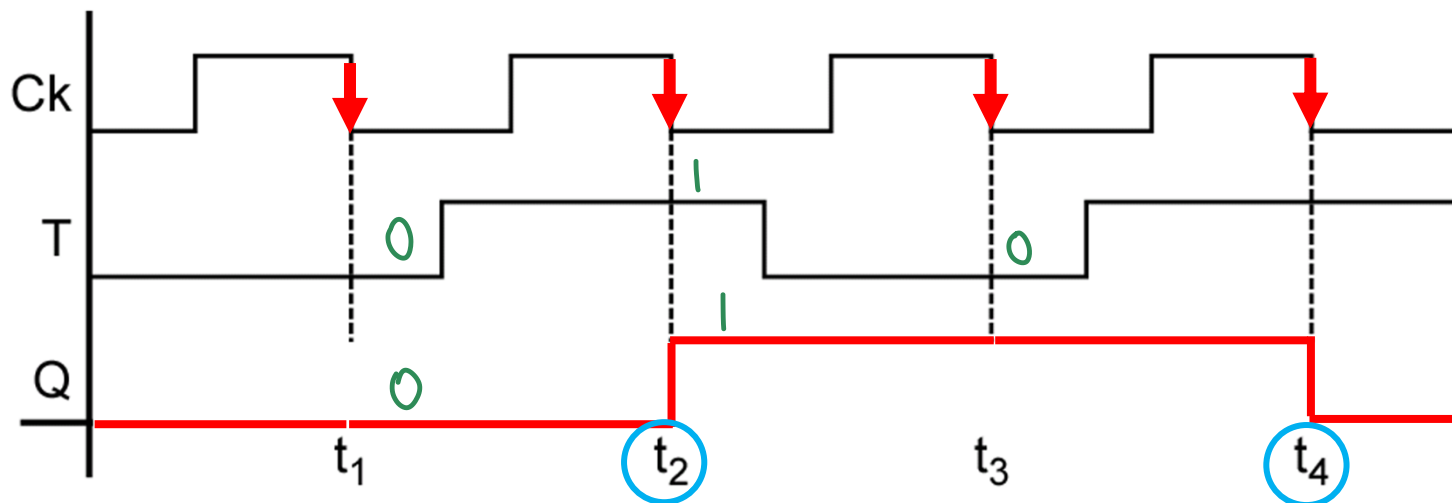
TRUTH TABLE

T	CLK ↑/↓	Output		Comments
		Q	Q _{next}	
0	↑/↓	0	0	HOLD
0	↑/↓	1	1	HOLD
1	↑/↓	0	1	TOGGLE
1	↑/↓	1	0	TOGGLE



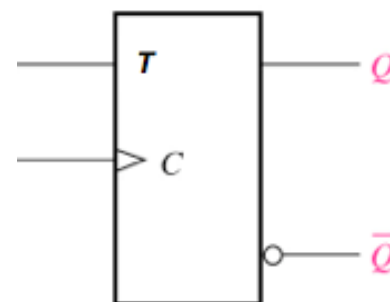
Logic symbol

Timing
Diagram for
T Flip-Flop
(Negative-
Edge
Triggered)

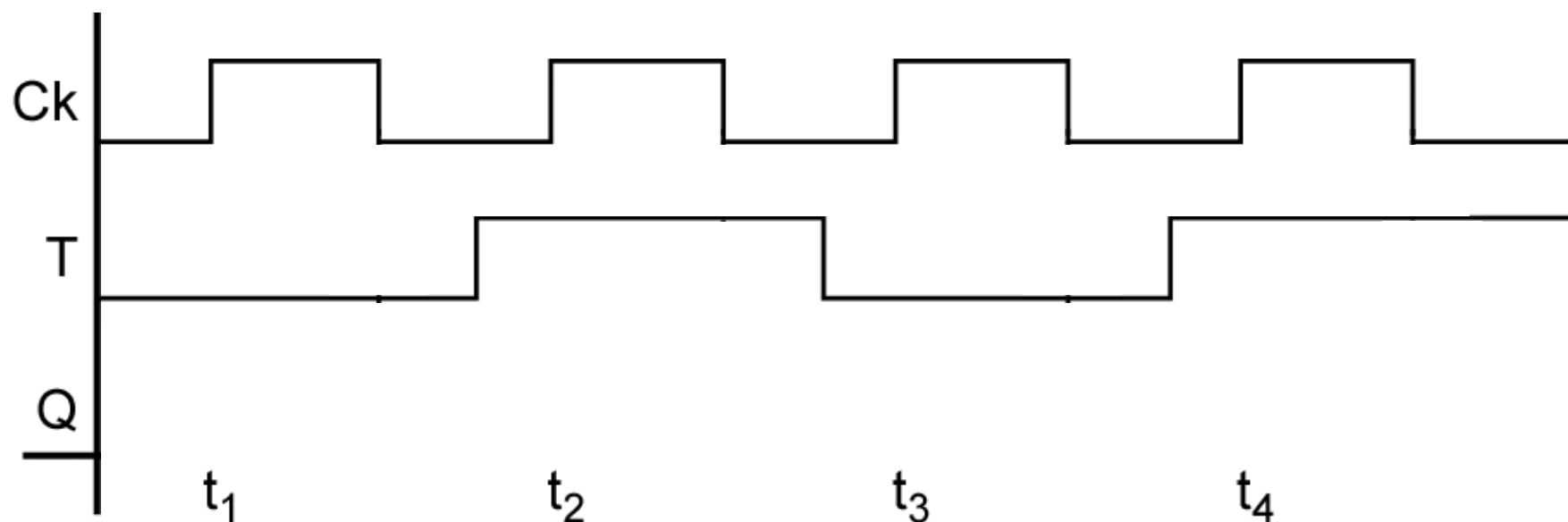


Try :

Consider a positive-edge trigger for a T-FF. Draw the waveforms for Q. Assume that the output is initially LOW.



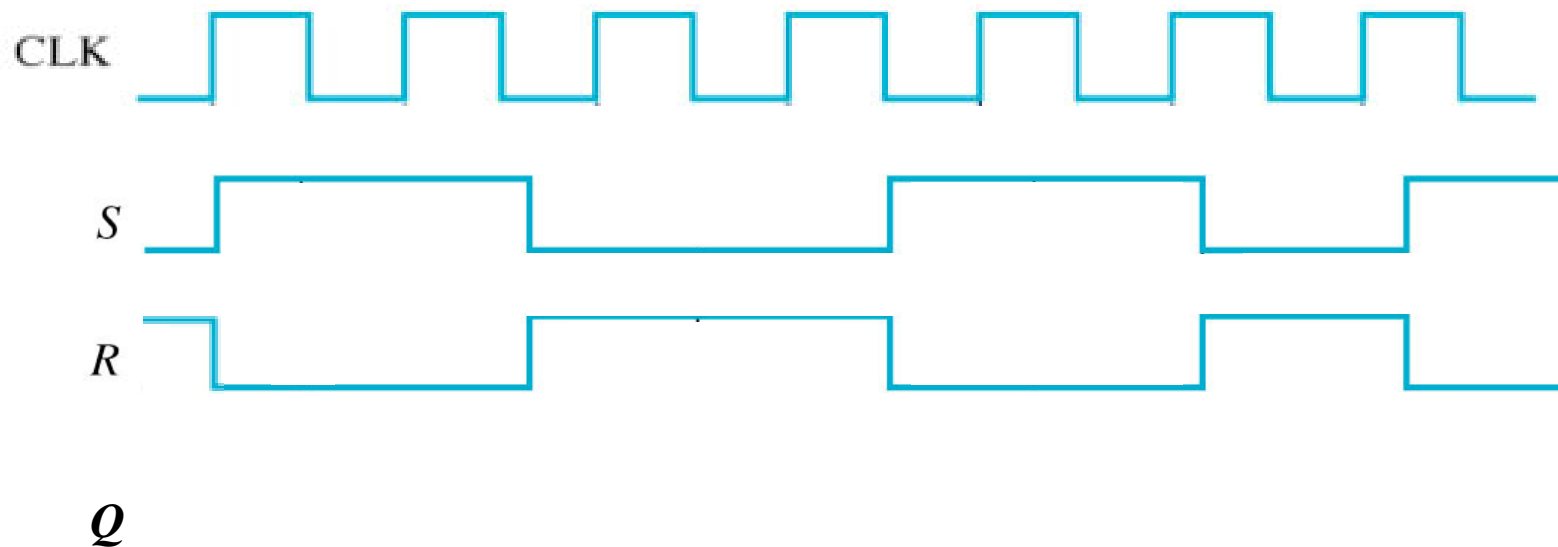
Logic symbol



Exercise 7.4:

Given the inputs S and R for flip-flop with the inputs always complement to each other and activated at negative edge-triggered.

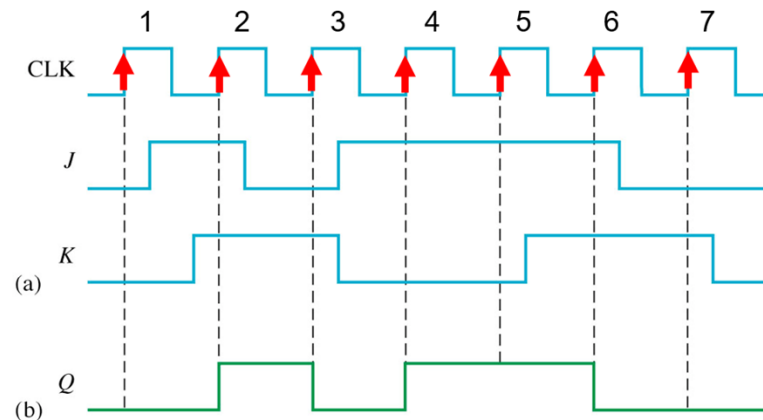
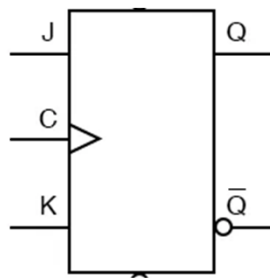
- (i) Draw the logic symbol for the flip-flop.
- (ii) Find the waveforms Q. Assume that the output is initially LOW.





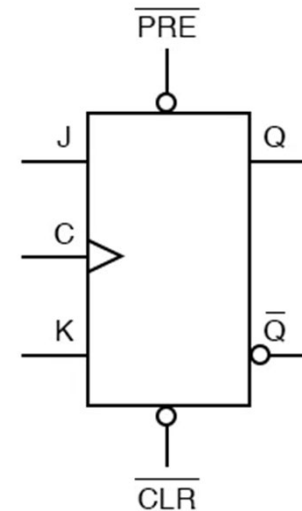
JK Flip Flop with Asynchronous Input

- The J and K inputs are called **synchronous inputs** since they only influence the state of the flip flop when the **clock** input is present.
- Synchronous $\rightarrow$ J and K inputs transferred on the triggering edge of the clock

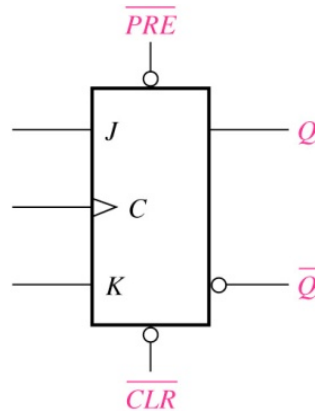


JK Flip Flop with Asynchronous Input

- This flip flop can also have **other inputs** called **Preset** and **Clear** that can be used for **setting** the flip flop to **1** or **resetting** it to **0**.
- Asynchronous → These inputs can **change the state** of the flip flop regardless of the clock signal.
 - Preset ($\overline{\text{PRE}}$) or Direct SET
 - Clear ($\overline{\text{CLR}}$) or Direct RESET



TRUTH TABLE



$\overline{PRE}$	$\overline{CLR}$	FF	MODE
0	1	SET	Asynchronous
1	0	RESET	Asynchronous
1	1	JK	Synchronous
0	0	-	-

FF Response:

$Q = 1$

$Q = 0$

FF Clock Working

Not Used

- Flip-flop input priority:

Priority	FF Input
Highest	$\overline{PRE}, \overline{CLR}$ (Asynchronous input)
Medium	Clock
Lowest	J-K, S-R, D, T (Synchronous input)

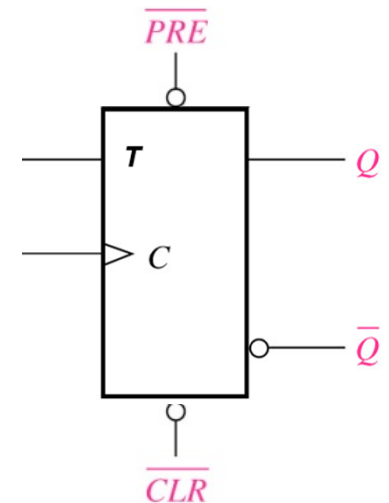
- Flip-flop asynchronous input has the highest priority.
 - Which means that if the asynchronous input active, the output will immediately change regardless the value of synchronous input or clock.

$\overline{PRE}$	$\overline{CLR}$	FF
0	1	SET
1	0	RESET
1	1	JK

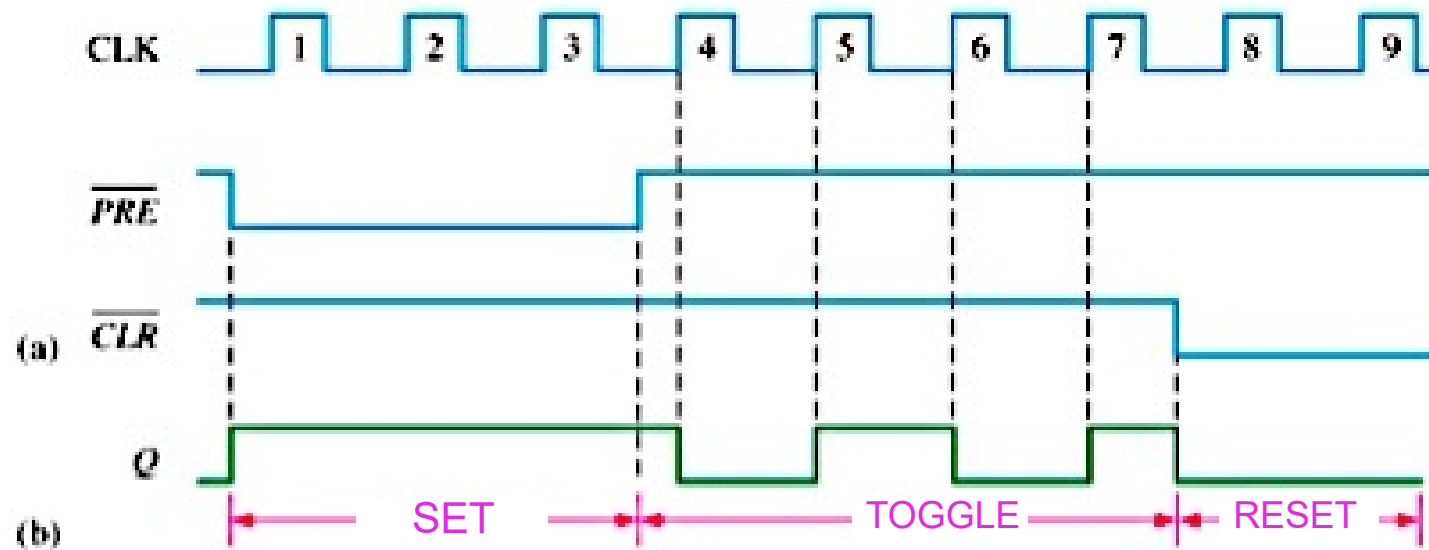
(Module: page 231)

Note:

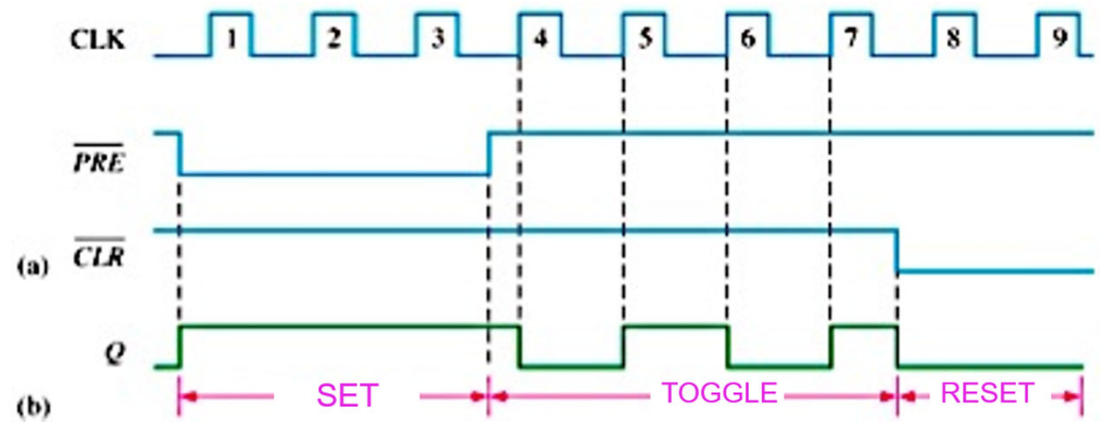
When $J = K = 1$,
it works as **T flip-flop**



Example: J-K Flip-Flop with both inputs are HIGH.
Find the waveform for Q.
Assume that Q is initially LOW.



$\overline{PRE}$	$\overline{CLR}$	FF	MODE
0	1	SET	Asynchronous
1	0	RESET	Asynchronous
1	1	JK	Synchronous

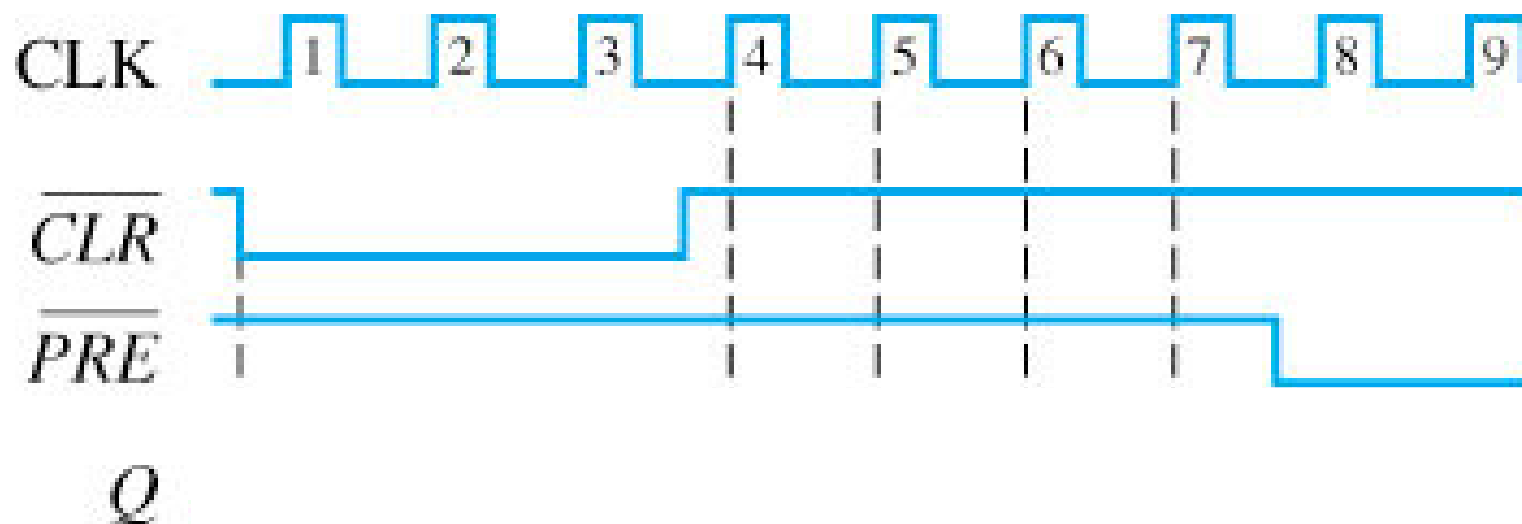


<i>clk pulse</i>	$\overline{PRE}$	$\overline{CLR}$	J	K	Q FF	Comment
1,2,3	0	1	1	1	SET	JK inputs - dont care
4,5,6,7	1	1	1	1	Toggle	Synchronous mode.
8,9	1	0	1	1	RESET	JK inputs - dont care

Exercise 7.5:

Find the waveform for Q.

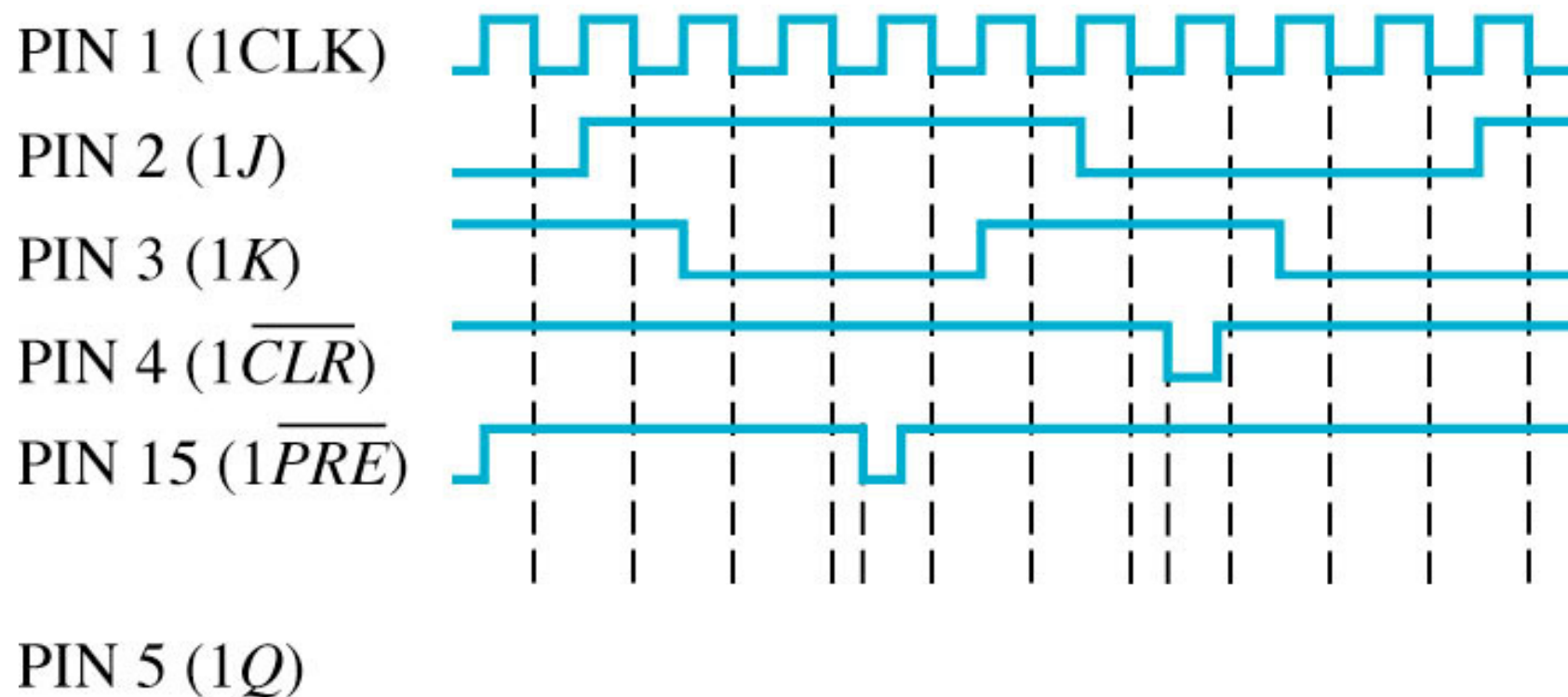
Assume that asynchronous J-K FF is used with both inputs are HIGH, and Q is initially LOW.



Exercise 7.6:

Find the waveform for Q for asynchronous J-K FF.

Assume that Q is initially HIGH.



Exercise 7.7: Complete the table in sequence.

Assume a J-K flip-flop is used with $\overline{PRE}$ and $\overline{CLR}$ inputs, and output Q.

$\overline{PRE}$	$\overline{CLR}$	J	K	Q	Mode	Comment
1	1	1	0			
1	1	1	1			
0	1	0	1			
1	1	0	0			
1	0	1	0			
0	1	1	1			
1	1	0	0			
1	1	1	1			

Mode: (Asynchronous / Synchronous)

Comment: (SET / RESET / HOLD / TOGGLED / PRESET / CLEAR / INVALID)



Summary

Latch & Flip-Flop

	Latch	Gated Latch	Flip-Flop
Output Characteristics	Output / next state depends to: • current inputs & • previous outputs / state	Output / next state depends to • enable gate input • current inputs • previous outputs / state	Output / next state depends to: • edge-triggered clock • current inputs • previous outputs / state

S-R Latch/FF

Block Diagram

Basic Operation Truth-table

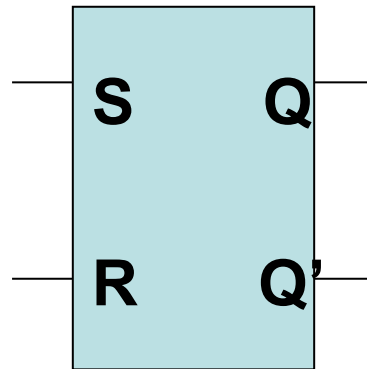
S	R	Q	Q'
0	0	Q	Q'
0	1	0	1
1	0	1	0
1	1	-	-

For:

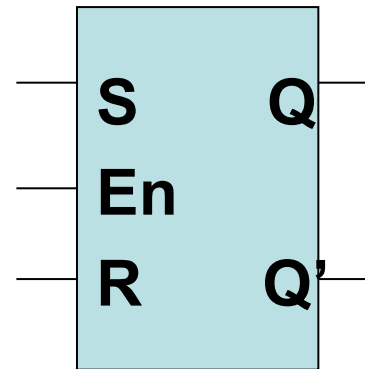
- En = 1 (Gated Latch)

-  &  FF

Latch

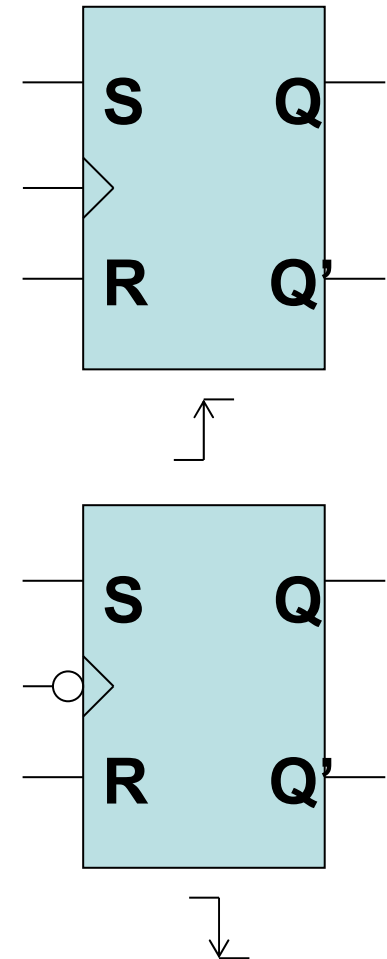


Gated Latch



S	R	En	Q	Q'
0	0	1	Q	Q'
0	1	1	0	1
1	0	1	1	0
1	1	1	-	-
X	X	0	Q	Q'

Flip-Flop



D Latch/FF

Block Diagram

Basic Operation Truth-table

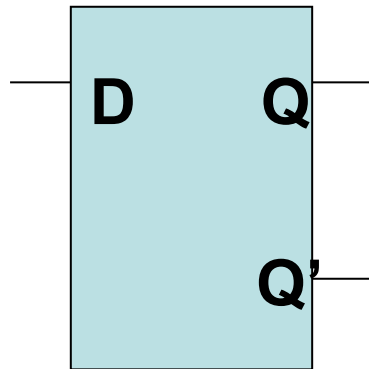
D	Q	Q'
0	0	1
1	1	0

Q follow D

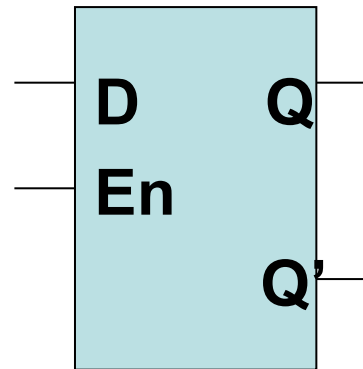
For:

- En = 1 (Gated Latch)
-  &  FF

Latch

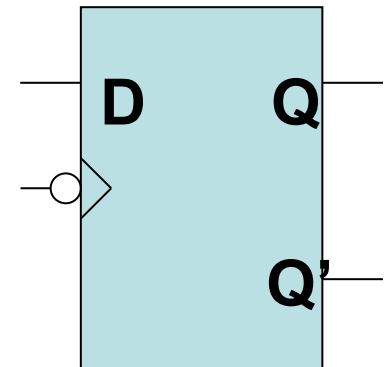
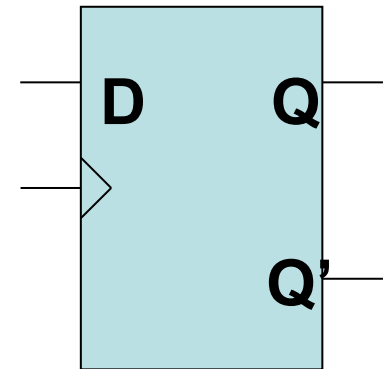


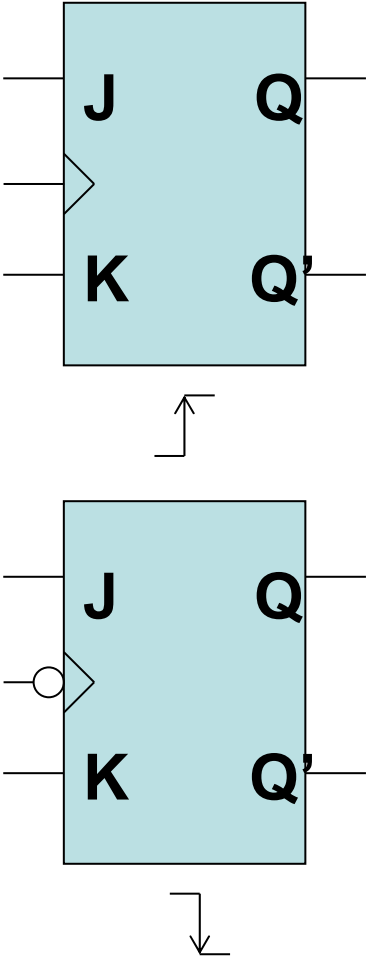
Gated Latch



D	En	Q	Q'
0	1	0	1
1	1	1	0
X	0	Q	Q'

Flip-Flop



J-K FF	Block Diagram																						
Basic Operation Truth-table	Latch	Gated Latch	Flip-Flop																				
<table border="1"> <thead> <tr> <th>J</th><th>K</th><th>Q</th><th>Q'</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>Q</td><td>Q'</td></tr> <tr> <td>0</td><td>1</td><td>0</td><td>1</td></tr> <tr> <td>1</td><td>0</td><td>1</td><td>0</td></tr> <tr> <td>1</td><td>1</td><td>Q'</td><td>Q</td></tr> </tbody> </table>	J	K	Q	Q'	0	0	Q	Q'	0	1	0	1	1	0	1	0	1	1	Q'	Q	Nil	Nil	
J	K	Q	Q'																				
0	0	Q	Q'																				
0	1	0	1																				
1	0	1	0																				
1	1	Q'	Q																				

Asyn. J-K FF

Block Diagram

Basic Operation Truth-table

Latch

Gated Latch

Flip-Flop

$\overline{PRE}$	$\overline{CLR}$	FF
0	1	SET
1	0	RESET
1	1	JK

J	K	Q	Q'
0	0	Q	Q'
0	1	0	1
1	0	1	0
1	1	Q'	Q

Nil

Nil

